MILITARY SPECIFICATION

PACKAGING OF MAIN PROPULSION SHAFTING, BEARINGS, BOAT AND SHIP PROPELLERS, AND ASSOCIATED REPAIR PARTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the cleaning, drying, preservation-packaging, packing and marking of main propulsion shafting, line shaft bearings, stern tube and strut bearings, boat and ship propellers (exclusive of controllable pitch propellers or shafting for controllable pitch propellers), and associated repair parts and tools. It also covers cleaning, drying, preservation-packaging, packing and marking of independently mounted main propulsion thrust bearings.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530 - Plywood, Flat Panel.
TT-W-571 - Wood Preservation; Treating Practices.
TT-W-572 - Wood Preservative; Water Repellent.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes; Paperboard Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-C-843 - Cushioning, Material Cellulosic.
PPP-C-1120 - Cushioning Material, Uncompressed Board Fiber for Packaging.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-75 - Tape, Preservation and Sealing.

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MIL-C-104 - Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-B-197 - Bearings, Anti-Friction, Associated Parts and Sub-Assemblies, Packaging of.
MIL-B-233 - Boxes, Repair Parts, Storage.
MIL-B-3106 - Board, Composition, Water-Resistant, Solid (Filler or Cushioning Pads).
MIL-C-3774 - Crates, Wood, Open, 12,000 and 16,000 Pound Capacity.
MIL-R-5001 - Rubber, Cellular Sheet, Molded and Hand Built Shapes; Latex Foam.
MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-C-6799 - Coatings, Sprayable, Strippable, Protective, Water Emulsion.
MIL-C-9897 - Crate, Slotted Angle, Steel or Aluminum, For Lightweight Airframe Components And Bulky Items (For Maximum Loads of 3000 Pounds).

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MIL-L-10547 - Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-B-13239 - Barrier Material, Waterproofed, Flexible, All Temperatures.
MIL-C-16173 - Corrosion Preventive Compound Solvent Cutback, Cold-Application.
MIL-R-20092 - Rubber Sheets and Molded Shapes, Cellular, Synthetic Open Cell (Foamed Latex).
MIL-F-22191 - Barrier Materials Transparent, Flexible, Heat Sealable.
MIL-C-23760 - Coating, Sprayable, Strippable, Protective, For Preservation, and Packaging of Weapons Systems and Components, Application Of.
MIL-C-26861 - Cushioning Material, Resilient Type, General.
MIL-C-46842 - Cushioning Material, Unicellular Polyethylene Foam (For Packaging Purposes).
MIL-T-50036 - Talc, Technical, Tl.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-731 - Quality of Wood Members for Containers and Pallets.
MIL-STD-758 - Packaging Procedures for Submarine Repair Parts, Utilizing Transparent, Flexible, Heat Sealable Film.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, With Appropriate Test Methods.

DRAWING

NAVSHIPS

NY-9-800-B-2095021 - Propeller Air-Lift Frame.

(Application for copies should be addressed to Commander, Mare Island Naval Shipyard (Code 564B), Vallejo, California 94592.)

PUBLICATION

NAVSHIPS

MANUAL 0900-028-3010 - Manual for the Inspection, Packaging and Storage of Shipborne Hull, Mechanical, Electrical and Electronic Equipments.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS

Association of American Railroads Rules

(Application for copies should be addressed to the Association of American Railroads, 59 E. Van Buren Street, Chicago, Illinois 60606.)

DEPARTMENT OF COMMERCE

CS-35 - Hardwood Plywood.
PS-1 - Softwood Plywood, Construction and Industrial.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be examined and tested as specified in 4.2. Detailed test results of the rough handling tests, when performed (see 4.2.1.2.1), shall be forwarded to the procuring activity.

3.2 Materials. Materials shall be as specified herein, and in accordance with the applicable specification.

3.2.1 New materials.

3.2.1.1 The use of newly developed preservation-packaging materials and procedures is encouraged and recommended and will be permitted under the conditions specified herein, provided they are equal to or better than the specified materials and procedures without increased cost to the Government.

3.2.1.2 Where the material or procedure is not covered by a specific government document, the manufacturer or fabricator shall furnish to the procuring activity documented evidence that the proposed material or procedure is equal to or exceeds the requirements of the specification for the material or procedure. If after a review of the documented evidence, it is the opinion of the procuring activity that the proposed material or procedure meets or exceeds the criteria established for required materials or procedures, authorization for use will be granted.

3.3 Preservation-packaging. Preservation-packaging shall be level A or level C, as specified (see 6.2).

3.3.1 Level A. Cleaning, drying, preservatives, preservative application criteria, and methods of preservation (unit protection) shall be in accordance with MIL-P-116, and as specified herein. Requirements in table I are assigned by item; methods and submethods of unit protection are assigned on the basis of the type of unit protection most commonly required for a specific component or item. Except where otherwise specified herein or in the contract or order, the selection of the submethod under a particular method of unit protection is at the option of the contractor. The contractor may provide any additional protection he considers necessary.

Table I - Preservation-packaging.

Item	Cleaning process MIL-P-116	Preservatives MIL-P-116	Method of preservation MIL-P-116	Remarks
Bronze sleeves shrunk on shaft	C-1	None	III	
# Certification form for propeller				See 3.10
Finish machined		P-1 MIL-C-16173 grade 1		
Steel shaft surfaces	C-5		I	
Brass shafts	C-1	None	III	
Nickel copper alloy shafts	C-1	None	III	
Full molded rubber or laminated phenolic	C-1	None	III	
Line shaft and main thrust bearings	C-3	P-2 MIL-C-16173 grade 2	I	
Manuals	C-1	None	III	See 3.7

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Table I - Preservation-packaging (cont'd.).

Item	Cleaning process MIL-P-116	Preservatives MIL-P-116	Method of preservation MIL-P-116	Remarks
Miscellaneous finished Bronze, brass, and nickel-copper alloy castings (stern tube stuffing box and gland)	C-1	None	III	
Miscellaneous finished Forged and rolled brass, bronze and nickel-copper alloy parts (studs, rings and keys)	C-1	None	III	
Miscellaneous finished Forged and rolled steel parts (coupling bolts, nuts, washers, keys and coupling flanges)	C-1	P-19 MIL-C-16173, grade 4 or P-18	I IC	
Miscellaneous finished Steel and iron castings bulkhead stuffing box	C-1	P-19 MIL-C-16173, grade 4 or P-18	I IC	
Propellers		See 3.3.1.6		
# Ship	C-1	Method IB, Modified		
# Boat	C-1	I		
Roller and ball bearings				See MIL-B-197
Rough machined Steel shaft forgings	C-1	P-1 MIL-C-16173 grade 1	I	
Rubber covered shaft areas	C-1	Dust with talc (see MIL-T-50036)	8 oz canvas	See 3.3.1.5
Stern tube and strut bearing bushings	C-1	None	III	
Stern tube and strut bearing staves Laminated phenolic Rubber, brass backed	 C-1 C-1	 None None	 III III	
Tube from stern tube and strut bearings	C-1	None	III	

3.3.1.1 Disassembly and matchmarking.

3.3.1.1.1 Disassembly. Equipment disassembly shall be the minimum necessary to make accessible for cleaning and preservation of all machined or critical surfaces. Removal of accessories or projecting parts which will facilitate protection of the equipment from damage, pilferage and loss, or reduction of cube is permitted where such removal will not affect permanent settings or alignments and where the removed part can be readily reassembled at the installation site without the need for special tools or gauges. Removed hardware (bolts, nuts, pins, screws, washers, and others) shall be reinstalled in mating parts and secured to prevent their loss.

3.3.1.1.2 Matchmarking. Removed parts of the equipment shall be matchmarked to facilitate reassembly. Removed parts shall be tagged with cloth shipping tags. The tags shall be attached to each of the mating parts. The tags and printing thereon shall be resistant to water, oil, and fading.

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3.3.1.2 Accessories. In order to facilitate packaging, accessories such as thermometers, sight glasses and route (rte) connectors be individually packaged in accordance with the applicable component specification. Accessories shall be included with the complete equipment inside the shipping container.

3.3.1.3 Preservative application. Immediately after product cleaning and drying, surfaces of the equipment or part shall be treated with a preservative in accordance with table I, or as specified herein. During and after preservative application, the product shall be handled in such a manner as to produce a uniform protective film. The protective film shall remain untouched for a period of time to allow the film to set before wrapping and packing are undertaken. Preservative shall not be applied to items or surfaces which are vulnerable to damage by the preservative and as required by MIL-P-116. Excessive use of preservative shall be avoided.

3.3.1.3.1 Strippable compound. The strippable compound shall conform to MIL-C-6799, type II, Multi-Coat System. The coating shall be applied directly, except as otherwise specified herein, to the propeller. The finished coat system shall consist of three separately applied coats: black, grey, and a white top coat, having a 12 to 15 mil total dry film thickness. The material can be sprayed in passes to a 4 to 5 mil dry thickness per coat without sagging. The compound application shall be in accordance with MIL-C-23760. Prior to the compound application, propellers incorporating an air emitter system shall have the air emitter holes covered with tape (see 3.3.1.6.1). In addition, an acrylic or transparent protective coating or covering shall be applied over the propeller hub identification markings to maintain these markings in a fully visible and readily readable condition.

3.3.1.3.2 External metal surfaces. Except for propellers protected with a strippable coating and other items preserved as required by table I, preservatives for external metal surfaces shall conform to MIL-P-116. Use of type P-1 shall be limited to surfaces where it would not normally have to be removed to place the equipment in operation, or where its removal by scraping or solvent action would not damage the part or equipment.

3.3.1.3.3 Internal metal surfaces. Internal metal surfaces which normally are in contact with oil shall be coated with type P-2 preservative. Internal metal surfaces normally in contact with water shall be coated with either a type P-3, or grade 5 compound conforming to MIL-C-16173. The grade 5 shall be used when "Chemical boil-out" cannot be used for compound removal, or where removal by hot water or steam is desired. The shaft bore shall be coated with type P-1 preservative.

3.3.1.3.4 Finished machined surfaces. All external finished machined steel surfaces protected with a preservative (see table I), shall be wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, followed by an overwrap with type I, grade C, class 1 or 2, with class selection at the contractor's option.

3.3.1.4 Closure of openings. Except for the closure of openings as specified in MIL-C-23760 (see 3.3.1.3.1), for propellers and shaft bores, openings shall be sealed with tape conforming to class 1 of PPP-T-60, or waterproof barrier conforming to class E-2 or L-4 of PPP-B-1055 and secured with tape as specified herein. When the covering is vulnerable to damage (puncture, tearing, etc.), the covering shall be further protected by use of weather-resistant hard board, wood, plywood, or metal covers. Pipe and fitting ends may be sealed with metal or plastic caps or plugs.

3.3.1.5 Shafting. After application of the preservative, each end of hollow shafts shall be sealed to prevent entry of water and other foreign material as follows: finished shaft bores with steel plugs and unfinished shaft bores with wood plugs.

3.3.1.5.1 Shaft sleeves, covered (such as rubber and plastic) shafting areas and dimensioned machined surfaces such as keyways, threads, tapers and polished bearing areas shall be provided a preliminary wrap of greaseproof material conforming to MIL-B-121, type I, grade A, class 2. All covered surfaces shall be securely wrapped with a minimum of three layers of not less than 8-ounce canvas or equivalent. The above protected surfaces shall then be provided physical protection by either of the following procedures:

(a) Application of approximately 0.0179 ± 0.005 inch thick (26 gage) steel wrap secured by means of galvanized, flat, nailless, type steel bands or
(b) Application of nominal one by two inch wood strip lagging (as shown on figure 2) running the entire length of the protected surface.

Lagging strips shall be spaced a maximum of one inch apart and shall be secured in place by means of corrosion resistant or treated steel strapping which shall be nailed to each lagging strip with a 3/4-inch maximum length roofing nail, or strapping may be stapled to

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the lagging strip. Care shall be taken that the point of the nail or staple does not penetrate the inside surface of the lagging strips.

3.3.1.5.2 Finished machined surfaces such as shaft flange faces including peripheries shall be protected by means of a 2-inch nominal thick wood disk which shall be secured to the flange faces. A layer of barrier material conforming to MIL-B-121, type I, grade A, class 2, shall be placed between the wood disks and metal surfaces.

3.3.1.5.3 Shafting surfaces in direct contact with wood cradles shall be wrapped with a minimum of three thicknesses of barrier material conforming to MIL-B-121, type I, grade A, class 2. Threaded or machined shafting ends shall be protected as specified in 3.3.1.3.4.

3.3.1.6 Propellers. Ships' propellers shall be protected by the strippable compound method (see 3.3.1.3.1). Boat propellers shall be protected with a contact preservative conforming to type P-1 or P-19 of MIL-P-116.

3.3.1.6.1 Prior to the compound application and application of the blade edge protectors, ships' propellers incorporating an air emitter system shall have the air emitter openings covered with a minimum 3/4-inch wide pressure-sensitive tape conforming to PPP-T-75, type I.

3.3.1.6.2 Blade edge protectors. Blade edge protectors shall be applied over the protective coating and be applicable only to ships' propellers. The edge of each propeller blade shall be covered with a minimum of two thicknesses of not less than 8-ounce canvas or equal, over which shall be placed a preformed, galvanized metal edge protector. The edge guard shall be fabricated of sheet metal, 14 to 18 gage, 0.0747 to 0.0478 inch respectively, depending upon blade size. A minimum of two, but not exceeding four corrosion-resistant or treated steel straps shall be secured to the metal edge protector. Strapping shall be applied on both sides of each blade following the blade surface contour, and positioned to prevent the edge protector from loosening or becoming dislodged during handling, shipment and storage.

3.3.1.6.2.1 Each propeller shall be marked with the following:

"DO NOT REMOVE BLADE EDGE PROTECTORS UNTIL PROPELLER INSTALLATION IS COMPLETE"

This marking information shall be stenciled on the top (white) coating. Letters shall be a minimum of one inch in height. When propellers are shipped without the strippable coating, the marking information shall be placed on a greaseproof, waterproof tag of appropriate size and securely affixed to the edge protector, steel strap, or lifting eye bolt.

3.3.1.7 Bearings. Ball or roller bearings shall be preserved in accordance with MIL-B-197. Lubricating systems, sleeve bearings (except prelubricated bearings) and housings shall be flushed with type P-2 preservative and thoroughly drained. Care shall be taken to insure that all surfaces of sleeve bearings and journals receive a preservative coating. Sufficient thickness of barrier material conforming to type I, grade A, class 2 of MIL-B-121 shall be used between the upper half linings of split sleeve bearings and journal to prevent vertical movement of the shaft or rotor after the bearing caps have been seated and bolted in place. The barrier material shall be projected beyond the assembly to permit easy detection. Each journal so prepared shall be plainly marked with a suitable waterproof tag stating "BEFORE OPERATING MACHINE, REMOVE BARRIER MATERIAL BETWEEN THE JOURNAL AND UPPER HALF LINING."

3.3.1.8 Repair parts and tools (accompanying equipment or for stock). Repair parts and tools shall be cleaned, preserved and packaged in accordance with table I, except as specified herein. Unless otherwise specified (see 6.2), repair parts shall be packaged one part per unit package, except that all parts comprising a single set or assembly shall be packaged together. When unit packaged as a set, assembly, or quantities greater than one, each item shall be wrapped or cushioned to prevent direct surface contact with surface of adjacent parts.

3.3.1.8.1 Other repair parts. Repair parts not specifically covered in table I shall be cleaned, preserved and packaged as specified in the contract or order (see 6.2).

3.3.1.8.2 Submarine repair parts. Repair parts known to be for submarine use shall have special reduced cube packaging in accordance with MIL-STD-758.

3.3.1.8.3 Special tools. Special tools shall be cleaned, preserved, and packaged in accordance with MIL-P-116 as follows:

(a) Tools made of ferrous metal with non-critical surfaces shall be cleaned by process C-1; dried as required, coated with type P-2 or P-6 preservative

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and packaged in accordance with method I of MIL-P-116. In lieu of the greaseproof wrap required, tools may be wrapped in transparent flexible greaseproof film conforming to type II of MIL-F-22191.

(b) Tools made of ferrous metal with critical surfaces shall be cleaned by process C-5, dried as required, coated with type P-9 preservative, and packaged in accordance with method IC of MIL-P-116. Selection of the sub-method of preservation shall be at the suppliers' option. Transparent flexible greaseproof film conforming to type II of MIL-F-22191 may be used in lieu of wrapping material conforming to MIL-B-121 for a preliminary wrap.

(c) Tools fabricated completely of non-ferrous materials or ferrous materials that are plated or otherwise treated to resist corrosion shall be packaged by method III of MIL-P-116.

3.3.1.9 Unit and intermediate containers. Unit containers, except those required by MIL-P-116 for the applicable method of preservation, and intermediate containers shall be folding paperboard, fiberboard, paperboard, metal stayed or set-up paperboard boxes in accordance with PPP-B-566, PPP-B-636, PPP-B-665 or PPP-B-676 at the option of the contractor. The gross weight of paperboard boxes shall not exceed 10 pounds. Fiberboard boxes shall not exceed 20 pounds. Unless otherwise specified (see 6.2), box closure shall be as specified in the applicable box specification or appendix thereto. When the gross weight exceeds 20 pounds, unit and intermediate containers shall conform to the containers specified in 3.4 for the level specified subject to the limitations specified herein.

3.3.1.10 Repair parts boxes. Unless specifically required, repair parts boxes shall not be required. When required, on-board repair parts shall be packed in type M (metal) or type W (non-magnetic) repair parts boxes conforming to MIL-B-233, as specified (see 6.2).

3.3.1.11 Index listing of repair parts and tools. An index list of repair parts and tools shall be inserted in each shipping container containing repair parts. The list shall be inserted in the index list support located on the interior side of the repair parts box cover, or shall be suitably placed within the box for quick accessibility. The list shall give a complete itemized list of the container contents including stock numbers, nomenclatures, and quantities. The list shall be enclosed in a waterproof bag or shall be so treated as to be resistant to water, oil, and fading.

3.3.1.12 Cushioning materials. Equipment or parts shall be cushioned, as required, to prevent damage to the equipment or parts during handling and transit and to prevent puncture or tearing of barrier materials utilized in packaging. Excessive use of cushioning shall be avoided. The requirements of MIL-P-116 shall be considered when determining the quantity of cushioning material required. Loose excelsior, shredded newspaper and similar materials are unacceptable for interior packaging except when enclosed in a sealed waterproof barrier material. Cushioning materials used within the unit package shall conform to MIL-B-3106, MIL-R-5001, MIL-R-20092, MIL-R-6130, PPP-C-1120, MIL-C-26861, MIL-C-46842, or PPP-C-843. Other cushioning materials may be used provided they conform to the cushioning requirements of MIL-P-116 and are non-corrosive to the packaged item.

3.3.1.13 Barrier materials. When using barrier materials conforming to MIL-B-121, the precautions as follows shall apply:

(a) The greaseproof side of grade A material shall be in contact with the part or surface.

(b) Grade C material shall not be in contact with metal surfaces or be used as an intimate wrap.

3.3.2 Level C. Preservation-packaging of equipment, accessories, components, and repair parts shall afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation.

3.3.2.1 Propeller blade edges shall be protected (see 3.3.1.6.2). Air emitter holes shall be covered with a neutral material; if pressure sensitive tape is used, there shall be no adhesive transfer upon tape removal that will restrict or obstruct the air emitter holes. Marking as specified in 3.3.1.6.2.1 shall apply.

3.4 Packing. Packing shall be level A or C as specified (see 6.2).

3.4.1 Level A.

3.4.1.1 General requirements. Shipping containers shall be of similar construction, of uniform size, and of minimum cube and tare consistent with protection required. Special tools, when furnished, shall be packed with the equipment for which they are intended. Con-

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tents of the shipping container shall fit in such a manner that the unit packed forms a compact, nonshifting load. Wood, plywood, or wood-cleated boxes shall be modified by the addition of skids. Unless otherwise specified herein, application of skids shall be in accordance with PPP-B-601 or PPP-B-621. Except where otherwise specified herein, crates shall be used for the shipment of individual items exceeding the weight limitations of the wood, plywood, or wood-cleated boxes. Unless otherwise specified (see 6.2), or herein, selection of the shipping container shall be at the contractor's option. The reference packaging, packing, and car loading figures are intended for general guidance use. Container figures applicable to propellers will require modifications to accommodate propeller eye bolts. Wherever reference is made to wood, wood sheathing, or wood boxes, plywood sheathing (see 3.4.1.1.2) and plywood boxes in accordance with PPP-B-601 are acceptable alternates.

3.4.1.1.1 Anchoring, blocking, bracing, and cushioning. Except as otherwise specified herein, equipment, accessories, and repair parts shall be anchored, blocked, braced, and cushioned, as applicable, in accordance with MIL-STD-1186 and the appendix to the applicable container specification. Cushioning materials shall conform to materials covered in 3.3.1.12.

3.4.1.1.2 Plywood. Plywood sheathing including container tops (crates), shall conform to NN-P-530. Unless otherwise specified (see 6.2), plywood selection for container use shall be as follows:

3.4.1.1.2.1 Multiple trip containers (see 6.3.2). Multiple trip containers shall be in accordance with CS-35, group A, type I (exterior, exterior type), grade 3-4 of PS-1, group B, exterior type, minimum A-C grade veneer.

3.4.1.1.2.2 Single trip containers. Single trip containers shall conform to CS-35, group A, type II (interior, interior type), grade 3-4, or PS-1, interior type, standard with exterior glue, veneer grade C-D (plugged).

3.4.1.1.3 Shafting, 3-inches in diameter and over and propellers, 30-inches through 108-inches in diameter.

3.4.1.1.3.1 Wood and plywood containers.

3.4.1.1.3.1.1 Multiple trip containers. All containers including interior packing media shall be fabricated and assembled to certify re-use. For wood or plywood boxes, the box cover (sides, ends, and top) can be assembled and fastened with nails; however, the assembled cover shall be fastened to the container platform or base by use of screws or bolts as shown on figure 1. Additional guidance for the application, selection, and spacing requirements for fasteners is contained in the applicable container specification.

3.4.1.1.3.1.2 Preservative treatment. Unless otherwise specified (see 6.2), all multiple trip (re-usable) containers or all of the finished wood or plywood members/parts shall be preserved for a minimum of one minute in preservative conforming to composition A or B of TT-W-572. When containers are painted, preservation shall be accomplished prior to the painting application. Alternatively, shipping containers or all of the finished wood or plywood members or parts shall be pressure treated in accordance with TT-W-571, table III, using preservative concentrations, "for use under moderate weathering conditions". Fasteners (nails, bolts, screws, washers, nuts, and so forth) for containers treated in accordance with TT-W-571 shall be galvanized or corrosion treated.

3.4.1.2 Main propulsion shafting.

3.4.1.2.1 Three-inches in diameter and less. Shafting, three-inches in diameter and under shall be packed in wood boxes conforming to PPP-B-621, style 1, modified as shown on figure 2. Table II provides sizes of members for boxes depending on shaft diameter, shaft length, and shaft weight. Container sides, ends, top, and bottom shall be one piece or Linderman jointed and without end butt joints for lengths up to sixteen feet. For lengths over sixteen feet, end butt joints are acceptable. Butt joints shall be alternated and located only in the end one-third areas of the length.

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Table II - Size of members for wood containers, shafting not exceeding 3-inches in diameter.

Shafting			Member sizes			Maximum spacing of cradles
Maximum Diameter	Maximum Length	Maximum Weight	Sides	Ends	Cross skids	
Inches	Feet	Pounds	Inches	Inches	Inches	Inches
1-1/2	12	100	1 by 4	2-2 by 4	2 by 3	16
1-1/2	14	200	2 by 4	2-2 by 4	2 by 3	16
2	20	200	2 by 6	2-2 by 6	2 by 3	18
3	25	300	2 by 8	3-2 by 8	2 by 4	20
3	16	400	2 by 6	3-2 by 6	2 by 4	20
3	20	400	2 by 8	3-2 by 8	2 by 4	20

NOTES: 1. For tops and bottoms, nominal one-inch lumber shall be used. When the width of top and bottom is over 7-1/2 inches, one by two-inch battens spaced approximately three feet apart shall be used; or 3/4-inch plywood, end spliced when required shall be used in place of lumber.

2. All cradles shall be of two-inch material with 1/4-inch plywood backing when shafts exceed two-inch diameter.

3.4.1.2.2 Finished shafting and rough machined shaft forgings, over 3-inches in diameter. One shaft complete with accessory parts, where applicable, shall be cradled on a skid type base (see figures 3 and 4 and MIL-C-104) for construction guidance. Unless otherwise specified (see 6.2), table III shall apply for skid base size members. Shafts and accessory parts where applicable shall be so cradled, mounted, and secured on the skid base to prevent shaft deflection, shifting, and damage that may result from storage, handling, and shipment. Skids less than sixteen feet in length shall not be spliced. Laminating and splicing of skids shall be in accordance with MIL-C-104. End headers shall be placed flush with the skid ends and bolted in place. To protect the bottom of the flange, a nominal 2-inch thick wood member the width of the flange box shall be secured to the skids directly beneath the flange box.

Table III - Sizes of members in skid assemblies for shipment of propulsion shafting, over 3-inches in diameter.

Items	Sizes of members for shafts with diameters of			
	Greater than 3 inches but less than 8 inches	8 inches to less than 16 inches	16 to 24 inches	Over 24 inches
	Inches	Inches	Inches	Inches
Skids	4 by 4 or 2-2 by 4's (flat)	4 by 6 (flat) or 2-2 by 6's (flat)	4 by 8 (flat)	6 by 8 (flat)
End headers	4 by 4	4 by 6 (flat)	4 by 8 (flat)	4 by 8 (flat)
Cross skids[3/]	3 by 4 (flat)	4 by 4	4 by 4	4 by 4
Cradles[1/] (cutout)	2-4 by 6's to 4 by 8	2-4 by 8's to 4 by 12	2-4 by 12's to 4 by 16 laminate with 3/4-inch plywood	2-4 by 16's and larger laminate with 3/4-inch plywood
Cradle[2/] spacing (maximum)	20	36	2-1/2 times diameter	2 times diameter
Lagging	1 by 2	2 by 2	2 by 2	2 by 2

1/ Cradles shall be fabricated in two equal parts, the bottom and top halves cut out to fit the shaft snugly with allowance for preservation, wrapping, lagging or other protection. Laminated cradles are used for shafts exceeding 16 inches in diameter.

2/ The number and location of cradles should be adjusted so that they are located at the quarter points for sling lifting.

3/ Cross skids shall be located at each cradle and shall be bolted to the skids with the same bolts used to fasten the cradle.

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3.4.1.3 Propellers. Propellers, 40-pounds and over shall be individually packed and provided with lifting eye bolts to facilitate propeller handling, including placement and removal from its shipping and storage container. Each propeller shall be marked or tagged with the following information "HANDLE BY EYE BOLTS". Marking information shall be applied in accordance with 3.3.1.6.2.1.

3.4.1.3.1 Thirty inches in diameter and under. Propellers, 30-inches in diameter and under, shall be multiple or individually packed (see 6.2) as follows:

(a) Multiple pack. Propellers shall be packed in wood boxes (see figure 5) conforming to PPP-B-621, class 2, style 2 for type 3 load. All sheathing, ends and cleats shall be minimum 1-inch lumber. A diagonal cleat to the same size as the regular cleats shall be used on each end of the box. One intermediate set of inside cleats or battens of not less than nominal 1 by 4-inch lumber shall be nailed and clinched to the sides, top and bottom midway between the ends. Hubs of propellers shall be fitted with suitable wood cones with drilled centers to fit a suitable tie bolt, on which propellers shall be strung. Tie bolts shall be securely suspended between the ends of the box by use of nominal 2 by 10-inch members securely cleated to the inside of the box ends. The size of the box shall be such as to insure not less than approximately 1 inch clearance on top, sides and ends. Boxes over 48 inches long shall be provided with an intermediate saddle midway between the ends which will firmly support the tie bolt from all sides. The saddle support shall be securely attached to the sides, top and bottom of the box. Suitable blocking which will not injure or mar the blades shall be provided to prevent propellers from turning on the tie bolt. Boxes shall be provided with skids in accordance with the applicable box specification. The gross weight of each box shall not exceed approximately 600 pounds.

(b) Individual packs.

(1) Propellers shall be packed in wood boxes conforming to the requirements of PPP-B-621, class 2, style 2 (reinforced), type 3 load as shown on figure 6. Blocking shall be cut from nominal 2 by 6 or 2 by 8-inch lumber with a minimum 10-inch length. Blocking shall be cut in pairs with the angle of cut conforming to the blade angle as shown on figure 7. Blocks shall be covered with at least three thicknesses of type I, grade A, class 2 material conforming to MIL-B-121 at areas of contact with the blades.

(2) When specified (see 6.2), propellers shall be individually packed in corrugated fiberboard boxes conforming to PPP-B-636, class weather-resistant, full overlap style as shown on figure 8. Blocking, bracing, and fillers shall be of the same material as the box except liners at blade contact area shall be solid fiberboard. Fillers, as required, shall be used under and over the hub areas and cut to fit snugly within the blocking sleeve. Propeller edges and finished faces of the hub shall be covered with barrier material conforming to type I, grade A, class 2 of MIL-B-121. Interior blocking, bracing, and fillers as shown on figure 8 shall be used for two and three-bladed propellers. Method V requirements of the appendix to the fiberboard box specification shall apply for box closure, waterproofing, and reinforcing.

3.4.1.3.2 Thirty to 60-inch diameter. Propellers shall be individually packed in wood boxes conforming to PPP-B-621, class 2, style 2 (reinforced), type 3 load as shown on figure 9. Joists shall span the box width. Table IV provides guidance for joist sizes, as required, for various loads and spans. The joist supports shall be at least 2-inches wider than the depth of the joists, e.g., 2- by 4-inch joists require a 2- by 6-inch support.

Table IV - Sizes of double joists required to support propellers in nailed wood box.

Weight of propeller	Size of joists[1/] required for span of					
	30 inches	36 inches	42 inches	48 inches	54 inches	60 inches
Pounds	Inches	Inches	Inches	Inches	Inches	Inches
Up to 400	2- 2 by 4	2- 2 by 4	2- 2 by 4	2- 2 by 4	2- 2 by 4	2- 2 by 6
400 to 600	2- 2 by 4	2- 2 by 4	2- 2 by 4	2- 2 by 6	2- 2 by 6	2- 2 by 6

[1/] All joists are used in pairs as shown.

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Propellers exceeding 600-pounds shall be packed in a container consisting of a skid type base and a container cover as shown on figure 1. The skid base and member sizes shall conform to MIL-C-104. The container cover shall be a modified class 2, style 2 (reinforced), type 3 load box conforming to PPP-B-621 consisting of ends, sides, and a top. The ends of the cover shall become the sides and be fastened to the base skids. The sides shall become the ends and be fastened to the base headers. The cover ends (sides) shall be reinforced on the interior with nominal 1- by 4-inch battens and diagonals. A bolt spacing member supported by side bracing members of not less than nominal 2 by 10-inch lumber and extending the full width of the box shall be nailed to the top and bottom of the box, midway between the ends. The hub of the propeller shall be fitted with a suitable wood cone with drilled centers to fit a suitable tie bolt on which the propeller shall be strung. The 3/4 inch tie bolt shall be suspended between the 2 by 10-inch bolt spacing members. A hole shall be drilled in the sheathing, top, and bottom to accommodate the bolt head and nut, so that counter-sinking will not be necessary in the bolt spacing member. The size of the box shall be such as to insure not less than approximately 2 inches clearance on top, sides and ends. Suitable blocking which will not injure or mar the blades shall be provided to prevent propeller from turning on the tie bolt. Boxes shall be strapped in accordance with PPP-B-621.

3.4.1.3.3 Sixty to 108-inches in diameter. Propellers, 60- to 108-inches in diameter shall be individually packed in an open type crate with a skid type base as shown on figures 10, 11, and 12. Container (cover and base) member sizes, fastening, and assembly shall be in accordance with the requirements of MIL-C-3774, MIL-C-104, or PPP-C-650 as applicable. The crate size shall be such as to allow approximately 2-inches of clearance between the propeller and the interior surfaces of the container top, sides, and ends. The maximum size propeller packed in a crate shall not exceed the common carriers dimensions for such cargo. When crates exceed rail shipment limitations, propellers shall be shipped uncrated and in accordance with the requirements for propellers exceeding 108-inches in diameter (see 3.4.1.3.4). The base of the crate shall be reinforced with at least 6- by 10-inch wood, load-bearing members. Blocking that will not mar or injure the blades, protective coating, and edge guards shall be provided to prevent propeller movement and shifting within the crate during handling, shipment, and storage.

3.4.1.3.4 Propellers exceeding 108-inches in diameter.

3.4.1.3.4.1 Rail shipment. Figures 13, 14, 15, 16, and 17 depict loading for rail shipments. Similar loading procedures may be used for barge shipments. Propellers shall be blocked with timber cribbing and topped by a rounded timber shaft which shall support the propeller as shown on figure 17. The wood members should be of 10- by 10- or 12- by 12-inch timber, group (species) III of IV of MIL-STD-731. The support member shall be covered to prevent propeller hub damage. Cleats shall be used to align and fasten the crib members together. Steel rods, threaded at each end shall be used to fasten the timbers to the car floor. Steel tie rods, minimum 1-inch diameter, shall be fastened to the timber shaft and to the car floor to provide longitudinal bracing. Blocking and bracing shall be applied between the crib walls to prevent side movement. To prevent propeller rotation (see figure 16), either blocking between the propeller blades and crib walls or holddown rods can be installed. The propeller blade(s) shall not be used to support the propeller weight or to block, brace, or prevent shifting of the propeller and shall not extend through an orifice or any opening in the car floor. Loading on open cars shall meet the requirements of the Association of American Railroads Rules governing car loading.

3.4.1.3.4.2 Truck shipments. The carrier vehicle shall be capable of transporting the propeller in a tilted position either on a separate shipping frame (see 3.4.1.3.4.3) or securely mounted on the carrier. Deviations to shipment by these modes shall be forwarded to Commander, Naval Ship Engineering Center, SEC 6200, Center Building, Prince Georges Center, Hyattsville, Maryland 20782, for evaluation and acceptance. Requests for a deviation shall include descriptive data including photographs of the proposed carrier and propeller loading procedure. The carrier shall contain a propeller centering and securing system consisting of, but not limited to, (a) a steel, threaded, centering shaft secured to the carrier; (b) a wood plug, clamps, or braces at the base of the propeller hub to center the propeller on the centering shaft; and (c) a metal centering plate of minimum 1/2-inch thickness, predrilled to accommodate the propeller centering shaft and lifting eye bolts (see 3.4.1.3). Metal plates shall be provided by the activity shipping the propeller. A non-metallic gasket type barrier shall be placed between the metal plate and the propeller hub. Chains or other metallic holddown devices that are added, when required for shipment and safety purposes, shall be covered with a heavy duty covering such as a fire hose. Holddown devices shall be arranged and secured to the carrier and propeller in a manner so that no contact with the propeller blade and blade edges shall ensue during shipment.

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3.4.1.3.4.3 Propeller shipping frame. Five or seven blade propellers, not exceeding 15 feet in diameter, may be shipped on metal frames as shown on figure 18. NAVSHIPS Dwg. NY-9-800-B-2095021 and instructions thereon can be used for construction and propeller mounting guidance.

3.4.1.3.4.4 Storage packs. Each propeller shall be placed on wood skids of sufficient size to support the propeller above the ground level. Alternatively, each propeller may be supported on a skid type base. In addition, each propeller shall be provided with an open-type cover consisting of sides, ends, and top. The cover may be of the nailed or demountable type, and of wood or metal construction. A clearance of not less than two inches should be allowed between the propeller and the closest member of the container cover. When in open storage, a flexible, waterproof cover should be placed over the container. The following specifications may be used for guidance for material selection, fabrication, and assembly of the skid type base and cover:

Base - MIL-C-104
Cover - MIL-C-3774, PPP-C-650, or MIL-C-9897

3.4.1.4 Stern tube and strut bearing.

3.4.1.4.1 Bearings or bearing staves shall be packed in wood or wood-cleated plywood boxes conforming to PPP-B-621, class 2, or PPP-B-601, overseas type, respectively, at the supplier's option.

3.4.1.4.2 Full molded or stave bearings not in excess of 90-pounds need not be strapped and shall be packed in wood-cleated fiberboard boxes conforming to PPP-B-591, class 2.

3.4.1.4.3 Staves in excess of 1000-pounds shall be packed in sheathed or covered crates conforming to MIL-C-104, type I, class 1 or 2, or PPP-C-650. Unless otherwise specified (see 6.2), selection of the crate type and style shall be at the supplier's option.

3.4.1.5 Line shaft and thrust bearings. Oil and grease lubricated line shaft and thrust bearings of the conventional roller, ball, or babbitt type shall be packed in containers as specified in 3.4.1.4.1.

3.4.1.6 Associated repair parts. Associated repair parts shall be packed in containers conforming to any one of the following specifications at the option of the contractor.

Specification	Container	Type or class
PPP-B-636	Fiberboard	Weather-resistant
PPP-B-640	Fiberboard, corrugated, triple wall	Class 2
PPP-B-585	Wood-wirebound	Class III
PPP-B-591	Fiberboard, wood-cleated	Overseas type
PPP-B-621	Wood, nailed and lock corner	Overseas, class 2
PPP-B-650	Crates, wood, open and covered	
MIL-C-104	Crates, wood, lumber and plywood sheathed	
MIL-C-3774	Crates, wood open	

Shipping containers, except PPP-B-636 fiberboard boxes, shall be closed and reinforced in accordance with the applicable container specification or appendix thereto. PPP-B-636 fiberboard boxes shall be closed, waterproofed, and reinforced in accordance with method V, appendix to PPP-B-636. The gross weight of wood, or wood-cleated boxes shall not exceed # 200-pounds, except when a single item exceeds 200-pounds. Fiberboard boxes shall not exceed the weight limitations of the applicable fiberboard box specification. Shipping containers, except crates and PPP-B-636 fiberboard boxes, shall have case liners conforming to MIL-L-10547. Caseliners shall be closed in accordance with MIL-L-10547. Barrier material conforming to MIL-B-13239 may be used to fabricate the caseliner. When boxes are packed with items or interior packages meeting the following requirements, caseliners are not required (see 6.2):

(a) Items which are completely painted and have no unprotected critical surfaces.
(b) Large items which are completely coated with preservative type P-1 or P-19, with critical interior item surfaces preserved and all openings sealed with authorized material.
(c) Method IC or IA packages.
(d) Interior packages which conform to weather-resistant types or class and are waterproofed in accordance with the applicable interior package container specification.

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3.4.1.6.1 Repair parts boxes. Repair parts boxes conforming to MIL-B-233 (see 3.3.1.10) shall be overpacked in containers specified in 3.4.1.6. The gross weight of parts shall not exceed 200 pounds, in any one box. Where the combined weight of a set exceeds 200 pounds, such parts shall be grouped and packed in two or more boxes numbered consecutively to show the number of boxes in a complete set. Unless otherwise specified (see 6.2), an exception will be made when an individual part weighs in excess of 200 pounds, in which case the part shall be packed in an individual repair parts box.

3.4.2 Level C. The bearings, propellers, shafting and associated repair parts shall be packed in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

3.5 Marking. In addition to any special marking required (see 6.2), and herein, interior (unit and intermediate) packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

3.5.1 Special marking.

3.5.1.1 Shafting and propeller containers. Container markings shall include the following, "IF CONTAINER IS DAMAGED, NOTIFY COMMANDER, NAVAL SHIP ENGINEERING CENTER, SEC 6200, CENTER BLDG., PRINCE GEORGES CENTER, HYATTSVILLE, MARYLAND 20782. CONTAINER TO BE OPENED ONLY BY THE INSTALLATION ACTIVITY". For propeller containers, the following additional marking shall be applied:

"CAUTION: THIS SIDE UP
(with arrows pointing up)
STOW FLAT ON CONTAINER SKIDS."

Above marking shall be plainly stenciled and placed on two adjacent container sides.

3.5.1.2 Multiple trip containers. Multiple trip containers (see 3.4.1.1 and 6.3.2) shall be marked "REUSABLE". Instructions shall be provided for container disassembly and content removal. Instructions shall be enclosed in a heat sealed, waterproof envelope, pouch, or bag, and firmly affixed to the outside of the shipping container in a protected location (preferably between cleats and adjacent to the identification markings).

3.5.1.2.1 Center of balance. The center of balance of packed crates and boxes shall be indicated by a plainly marked vertical line on the lower portion of each container side, and the words "CENTER OF BALANCE" plainly stenciled. Letters shall be of not less than 1-3/4 inches in height.

3.5.1.2.2 Sling points. Sling points shall be legibly marked "SLING HERE".

3.5.1.2.3 Structural markings. Structural markings shall be applied in accordance with MIL-STD-129.

3.6 Commodity specifications. When an item is procured in conformance with a specification having detailed preservation-packaging, packing and marking requirements which differ from the requirements of this specification, the preparation for delivery requirements of the item specification shall apply.

3.7 Manuals. Manuals which accompany shipments shall be packaged in transparent, waterproof, plastic bags, minimum 4 mil thick and heat sealed. Manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable manual specification or packed in containers conforming to the requirements for the level specified.

3.8 Air shipment. Items required for movement by air shipment shall be handled for transportability in accordance with the procedures prescribed by the Military Airlift Command Air Base receiving the item for airlift.

3.9 Storage. Unless otherwise directed by the Naval Ship Engineering Center, propellers, propeller blades, fillet gages, and ships' propulsion shafting shall be placed in storage as prescribed by NAVSHIPS Manual 0900-028-3010, Appendix A.

3.10 Propeller certification form. The certification form shall be enclosed in a minimum 0.004 inch thickness transparent, waterproof, greaseproof bag, prominently marked "Certification Form" and the bag shall be heat sealed. The bag shall be attached or secured

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to the propeller hub or blade in a manner so that it is visible, readily available, and will remain with the propeller until propeller use. When propellers are boxed or crated, the bag may be firmly affixed to the outside of the shipping container in a protected location, preferably between the cleats on the end of the container adjacent to the identification markings.

3.11 Workmanship. Workmanship shall be such that, when the proper procedures are followed, materials and equipment being processed, cleaned, preserved-packaged, packed and marked, will receive protection against corrosion and deterioration during shipment and prolonged periods of storage, and will require minimum of reprocessing for service or further storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. There shall be two types of inspection in accordance with MIL-P-116, as follows:

(a) First article (see 4.2.1).
(b) Quality conformance (see 4.2.2).

4.2.1 First article.

4.2.1.1 First article inspection sample. The contractor shall conduct inspection on one complete package, packed for shipment, to ascertain that the cleaning, drying, preservation-packaging, packing, and marking of the equipment or item conforms to this specification. The first article sample will not be required when such a pack has previously been inspected and accepted for the same method for an identical or similar item by the same contractor and satisfactory evidence can be furnished to the Government that the equipment or items have been prepared identically with the previously appproved pack. First article inspection shall be repeated when changes in preservation-packaging and packing materials, or processes or designs are made.

4.2.1.2 First article examinations and tests. The sample first article pack of the equipment or item shall be subjected to the examinations and tests of MIL-P-116 and the tests as specified hereinafter. A first article test will not be required when one of the following applies:

(a) Level C packaging is specified.
(b) Level C packing is specified.
(c) Detailed packing instructions are furnished by the procuring activity.

4.2.1.2.1 Rough-handling tests. When specified (see 6.2), rough-handling tests shall be conducted. Tests shall be as specified in MIL-P-116. For packs exceeding 200 pounds (or 100 pounds where equipment is secured to the base of the container) the railroad car test method of 4.2.1.2.1.1 may be used in place of the pendulum-impact test or incline-impact test of MIL-P-116 at the option of the contractor.

4.2.1.2.1.1 Railroad car test method. The pack shall be securely blocked to prevent movement on the car. Anti-skid plates shall not be used for this test. The car shall be impacted at approximately 10 miles per hour.

4.2.2 Quality conformance inspection.

4.2.2.1 Sampling. Sample items, packages and packs shall be selected and inspected to determine conformance to the requirements of section 3 herein in accordance with MIL-P-116.

4.2.2.2 Examination.

4.2.2.2.1 Preservation-packaging. Examination of the preservation-packaging, and marking requirements not covered by referenced specifications shall be performed on the basis of the sampling procedures and inspection levels being used to determine examination requirements under MIL-P-116.

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4.2.2.2.2 Packing and marking. Examination of packing and marking requirements not covered by MIL-P-116 or any specification referenced herein shall be performed on sample packs selected in accordance with MIL-STD-105 at inspection level I. Any pack having one or more defects shall be cause for rejection. Lots shall be accepted or rejected in accordance with acceptable quality level (AQL) of 2.5 percent defective.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Intended use. The cleaning, drying, preservation-packaging, packing, and marking requirements specified herein are intended to insure proper and safe storage and transportation of equipments, associated repair parts, and tools for direct shipment to Government activities; shipments processed at a military activity or agency; as a reference in section 5, "Preparation for Delivery" of commodity specifications and standards, and for preparing preparation for delivery requirements in contracts or order.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packaging (see 3.3) and level of packing (see 3.4) required.
(c) Quantity of repair parts per unit package (see 3.3.1.8).
(d) Preservation-packaging requirements of repair parts not specifically covered (see 3.3.1.8.1).
(e) Box closure, if other than specified (see 3.3.1.9).
(f) Type of repair part boxes required, and if required (see 3.3.1.10).
(g) Level of packing required (see 3.4).
(h) Shipping container required, if other than supplier's option (see 3.4.1.1).
(i) Plywood selection, if other than specified (see 3.4.1.1.2).
(j) Wood preservative, if required (see 3.4.1.1.3.1.2).
(k) If table III applies to skid base size members (see 3.4.1.2.2).
(l) Required packing quantity (see 3.4.1.3.1).
(m) When fiberboard box is required (see 3.4.1.3.1 (b)(2)).
(n) Crate type and style, if other than supplier's option (see 3.4.1.4.3).
(o) When caseliners are required (see 3.4.1.6).
(p) Gross weight exception for repair parts boxes (see 3.4.1.6.1).
(q) Special marking, if required (see 3.5).
(r) Storage, if other than specified (see 3.9).
(s) When rough handling tests are required (see 4.2.1.2.1).

6.3 Definitions. Definitions of preservation-packaging, packing terms, including materials, processes, methods, and equipment will be found in Federal Standard No. 75, Glossary of Packaging Terms.

6.3.1 Levels of protection. To help determine the extent of preservation-packaging required to protect an item against specific hazards of storage, transportation, and handling, the Department of Defense has established three levels of protection whose main objectives are to provide uniform, efficient, and economical protection to supplies and equipment. The levels of protection are based upon the performance expected of the package. The performance criteria can be summarized as:

(a) Level A. The degree required for protection against the most severe conditions known or anticipated to be encountered during shipment, handling, and storage. Preservation-packaging and packing designated level A shall be designed for direct exposure to all extremes of climatic, terrain, operational, and transportation environments without protection other than that provided by the package and pack. The conditions to be considered include but are not limited to:
  (1) Multiple rough handling during transportation and intransit storage from manufacturer to ultimate user.
  (2) Shock, vibration and static loading during shipment, including deck shiploading and offshore or over-the-beach discharge to ultimate user.
  (3) Environmental exposure during transit where port and warehouse facilities are limited or nonexistent.
  (4) Extended unimproved open storage in all climatic zones, particularly while under static loads imposed by stacking.

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(5) Special package and pack features for field and combat operations (handling and utility).
(6) Special features are required by combat development agencies.

(b) Level C. The degree required for protection under known favorable conditions during shipment, handling and limited tenure of storage. Preservation-packaging and packing designated level C shall be designed to protect items against physical and environmental damage during known favorable conditions of shipment, handling and storage. In general, the following criteria will determine the requirements of level C.
(1) Limited handling during transportation and intransit storage.
(2) Limited shock, vibration and static loading during the transportation cycle.
(3) Controlled warehouse environment for temporary periods.
(4) Effects of environmental exposure during shipment and intransit delays.
(5) Stacking and supporting superimposed loads during shipment and temporary storage.
(6) Item characteristics require no special or peculiar preservation-packaging and/or packing provisions.

6.3.2 Multiple trip container. The design features are such that, subject to simple replacement of components, the container can be employed for more than one round trip from initial supply source to the consumer and return. A typical example is a demountable wooden box/crate equipped with fasteners permitting ready assembly, disassembly, and reassembly.

6.4 Detailed information. Supplemental information on preservation-packaging, and packing may be found in the following manuals:

DSAM4145.2 Vol. I, TM38-230-1, NAVAIR 15-01-1, AFP 71-4, Vol. I, MCO P4030.3
Preservation, Packaging and Packing of Military Supplies and Equipment
Preservation and Packaging (Volume I) (Federal Stock Number 0715-010-0290)

DSAM4145.2 Vol. II, TM38-230-2, NAVAIR 15-01-2, AFP 71-4, MCO P4030.21A
Preservation, Packaging and Packing of Military Supplies and Equipment
Packing (Volume II) (Federal Stock Number 0715-010-0280)

DSAM4145.7, TM38-236, NAVAIR 15-01-3, AFP 71-8, MCO P4030.30
Preparation of Freight for Air Shipment (Federal Stock Number 0715-010-270)

DSAM4145.3, TM38-236, NAVAIR 15-01-3, AFP 71-8, MCO P4030.30
Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft (Federal Stock Number 0715-010-0021)

(Copies of these manuals may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, (NPFC 105) 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

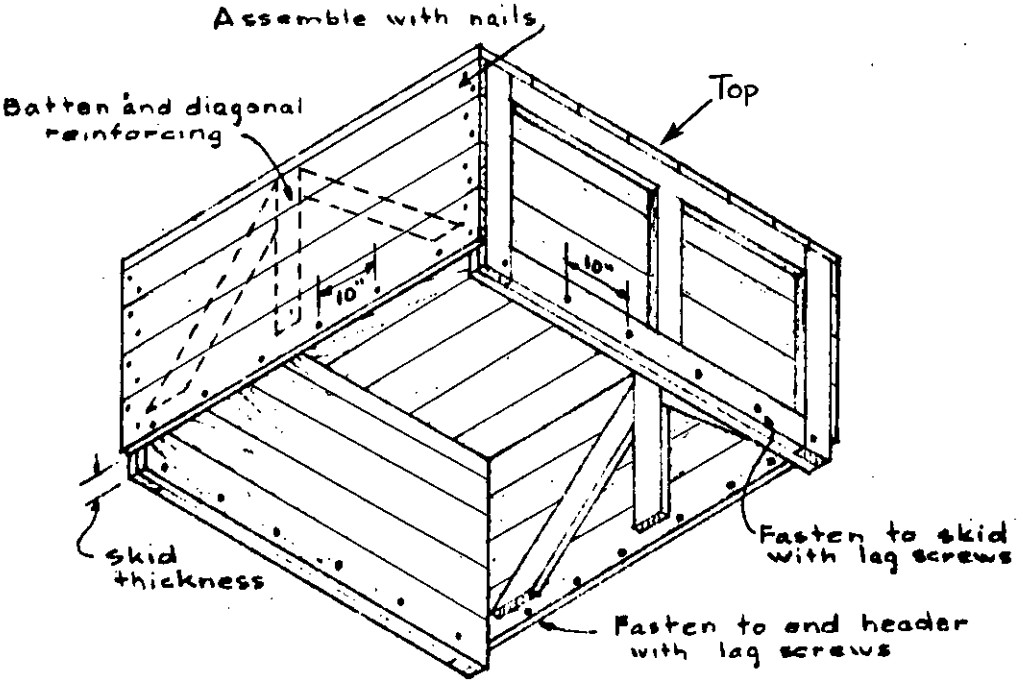
6.5 First article inspection.

6.5.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

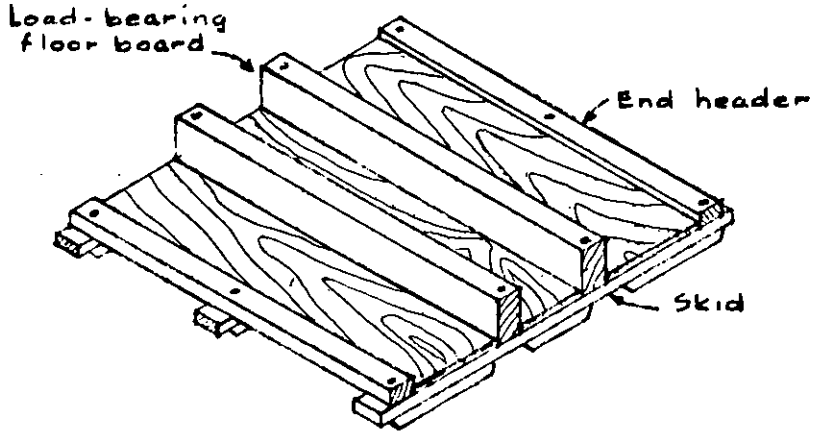
6.6 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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COVER



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BASE

Figure 1 - Cover and base (modified nailed-wood box) for propellers up to 60 inches in diameter.

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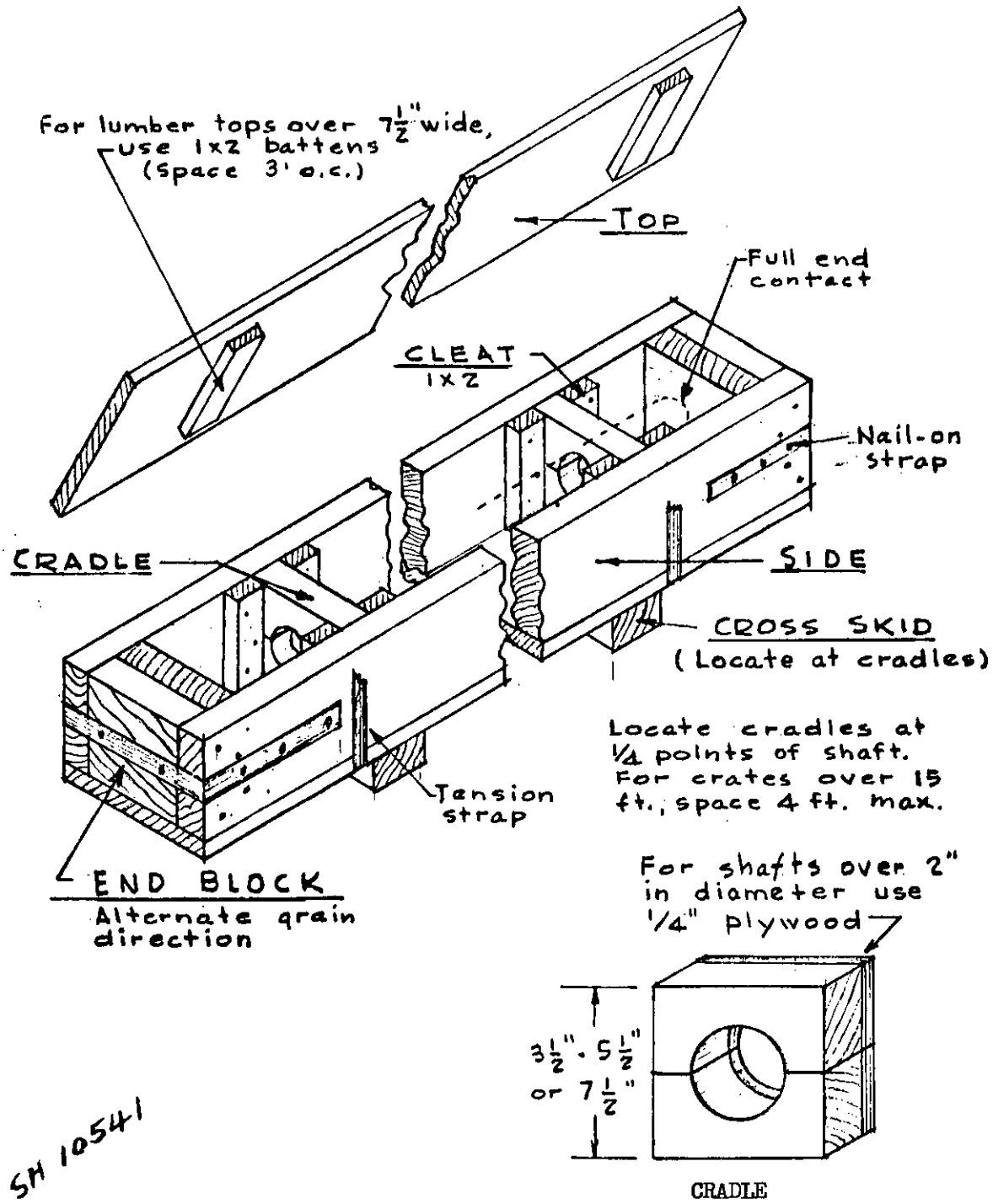


Figure 2 - Wood box for shafting under three inches in diameter.

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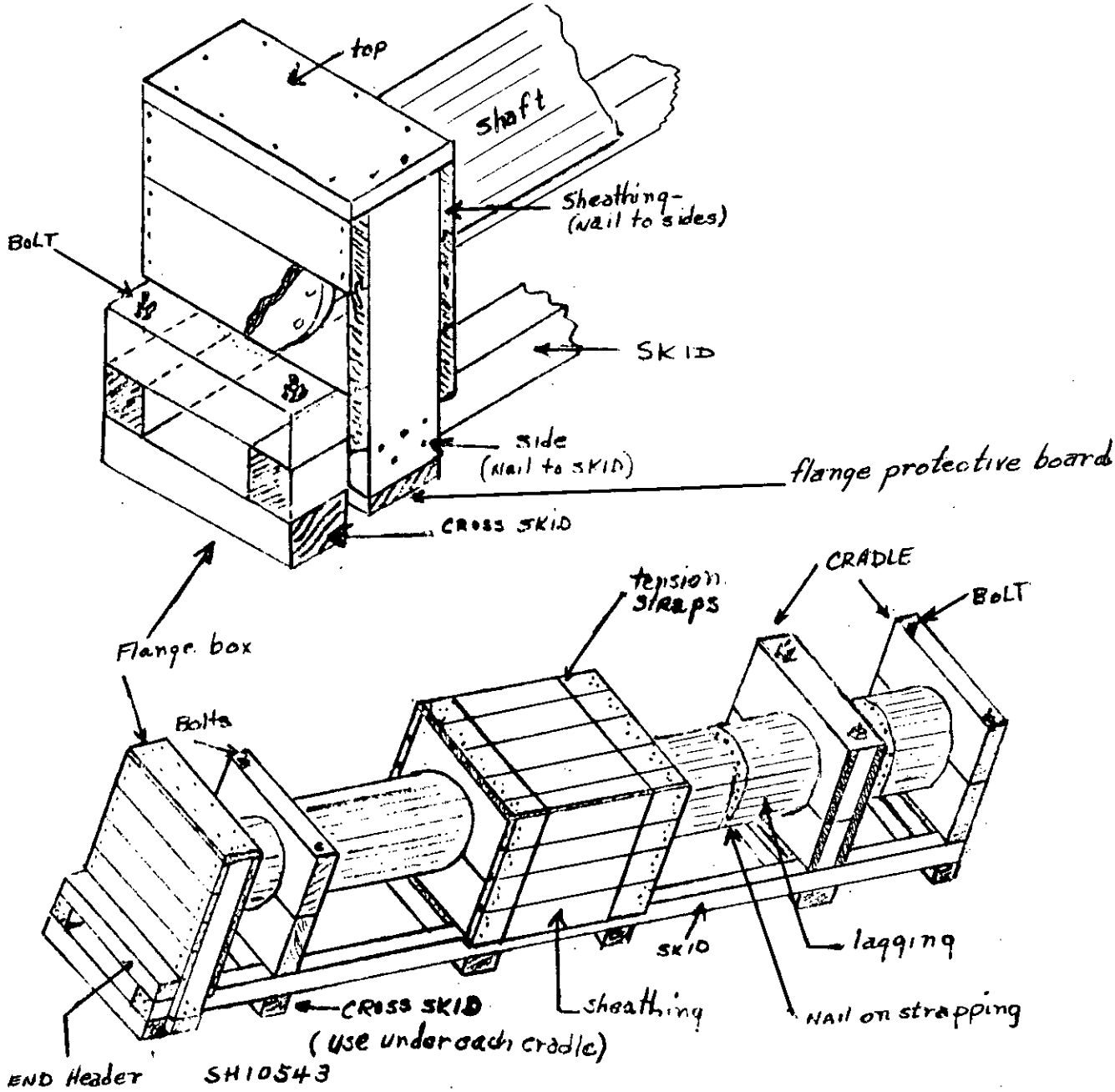
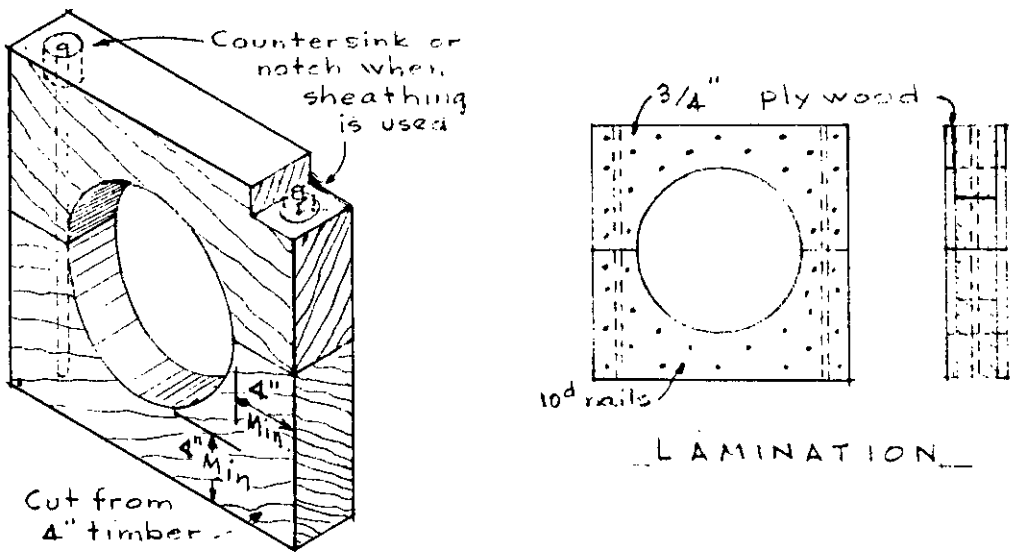
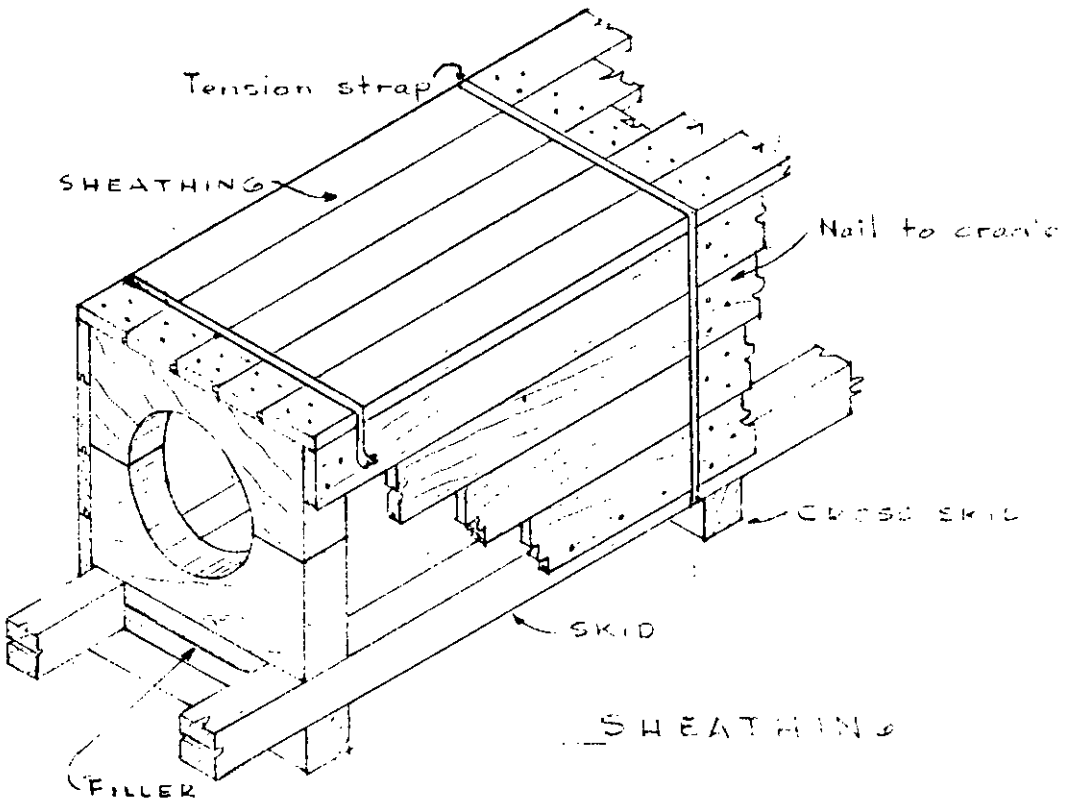


Figure 3 - Skid assembly (composite view) for shafts over 3-inches in diameter.

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Figure 4 - Skid assembly details.

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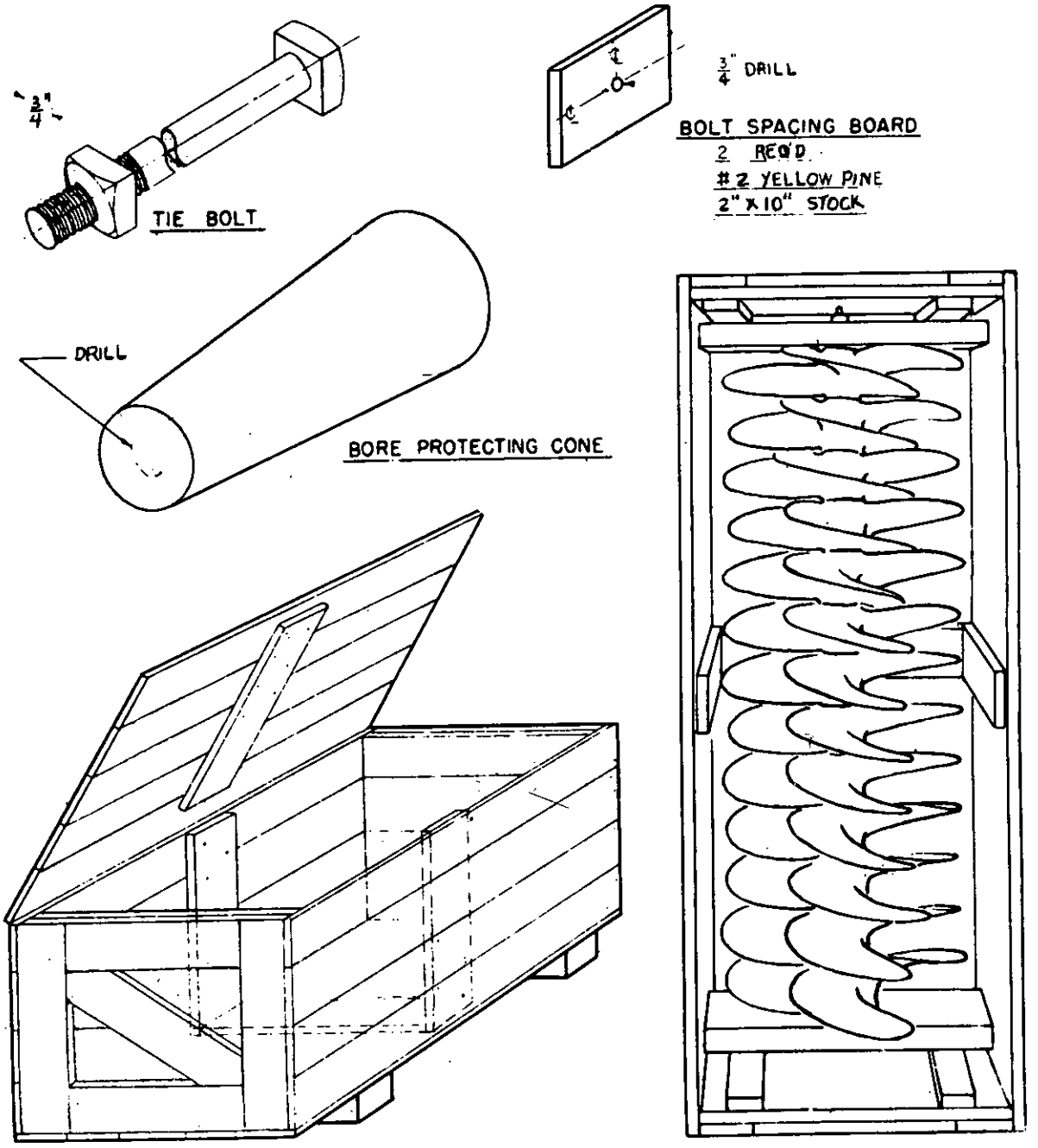
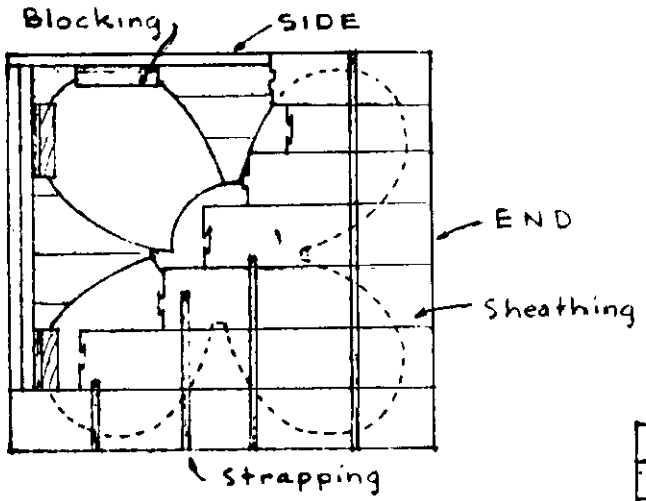


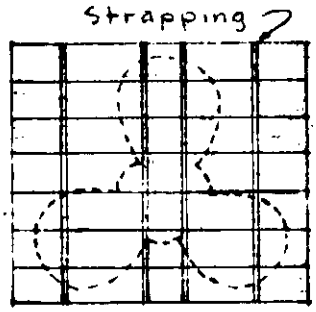
Figure 5 - Box for boat propellers 30 inches and under.

SH8943

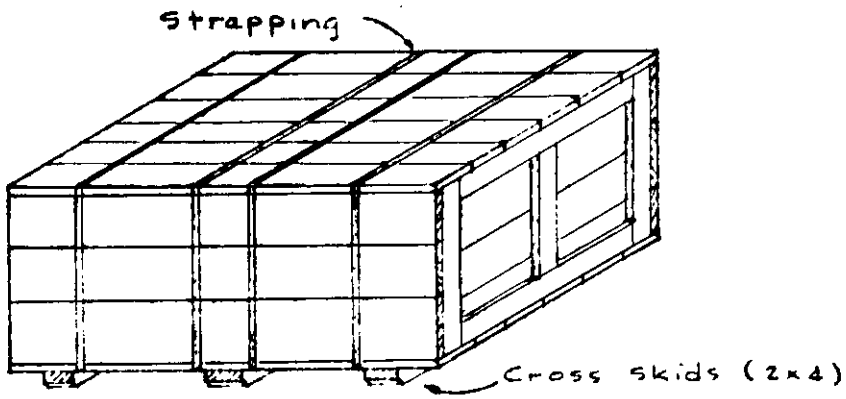
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TOP VIEW
(4-blade propeller)



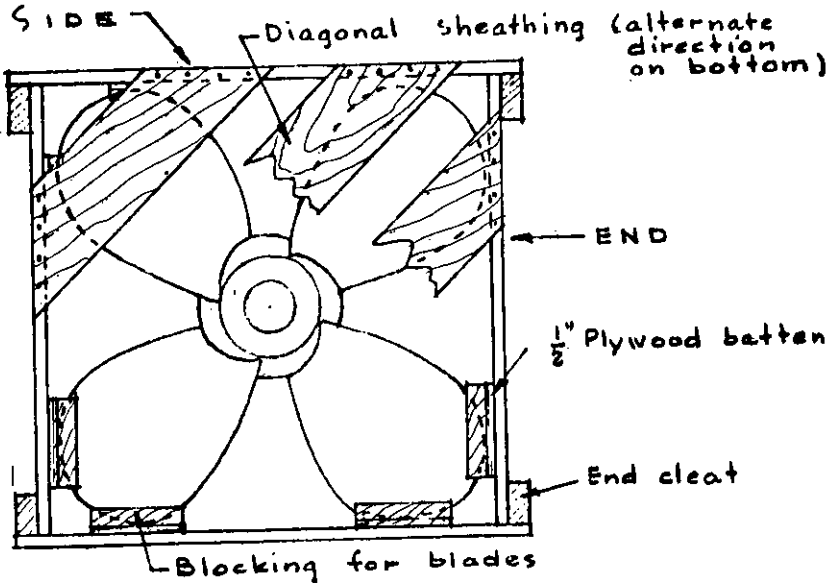
TOP VIEW
(3-blade propeller)

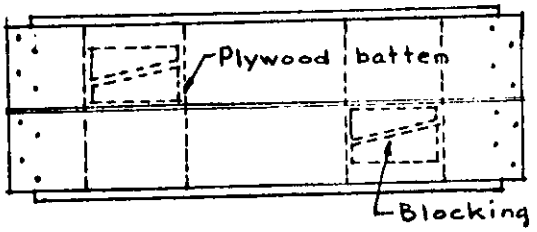


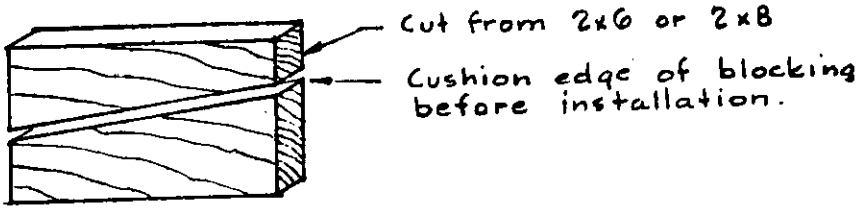
SH10544 HANDLING SKIDS

Figure 6 - Nailed wood box for propellers 30 inches in diameter and under.

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Figure 7 - Blocking for propellers 30 inches in diameter and under.

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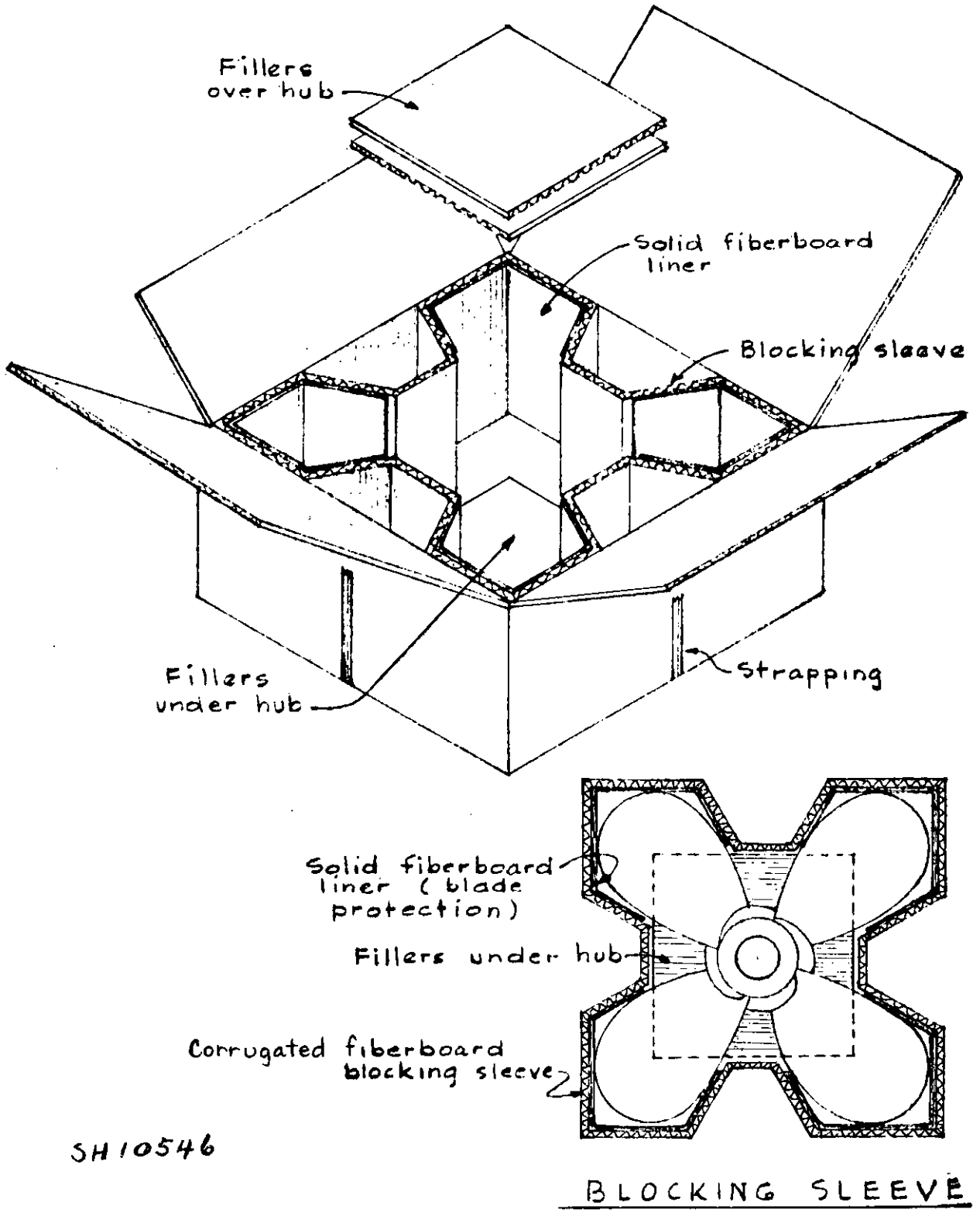
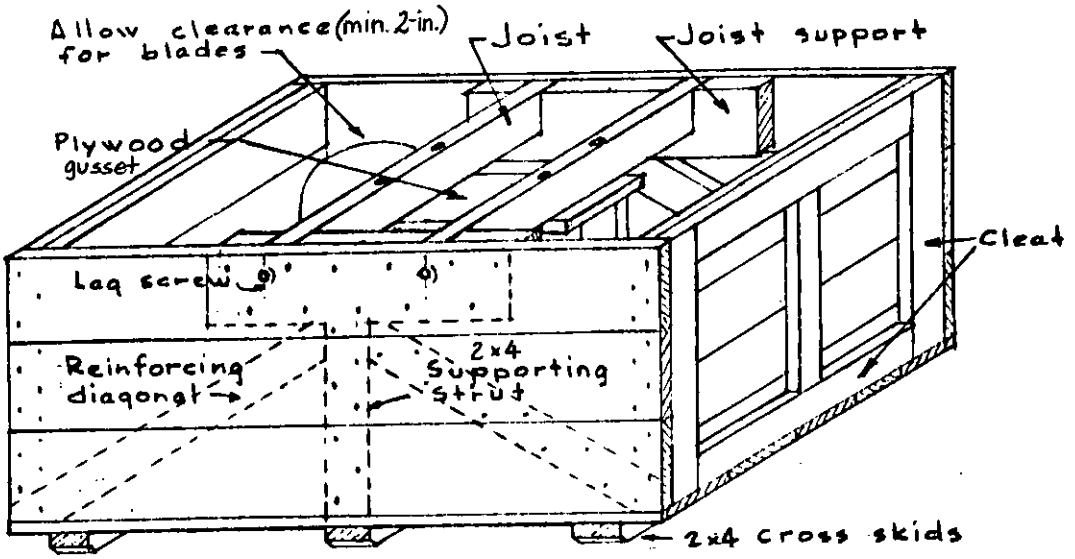
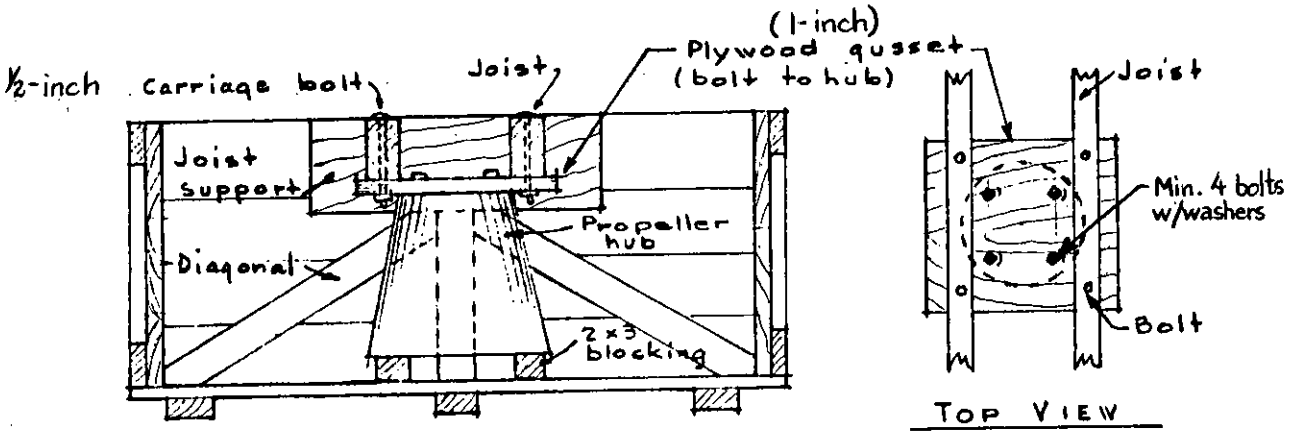


Figure 8 - Fiberboard pack for propellers 30 inches in diameter and under.

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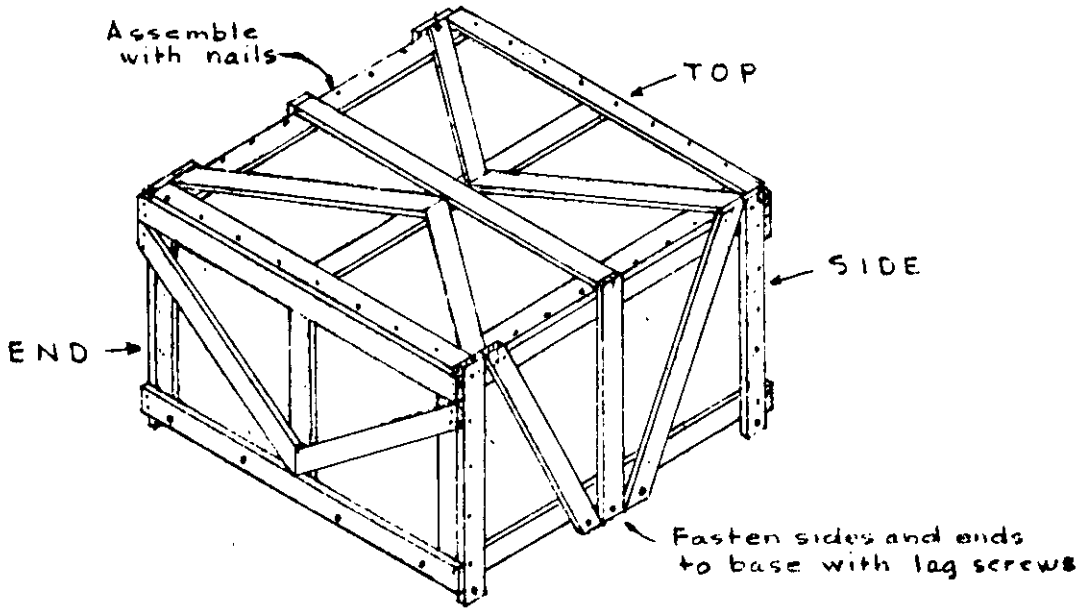


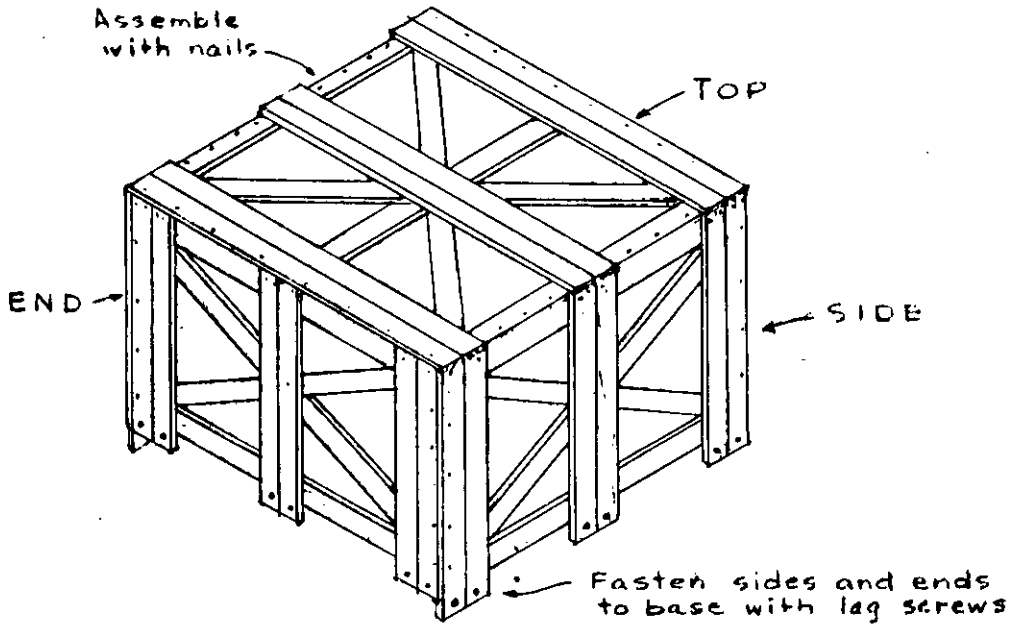


SH 10547

Figure 9 - Wood box for propellers 30 - 60 inches in diameter, not exceeding 600 pounds.

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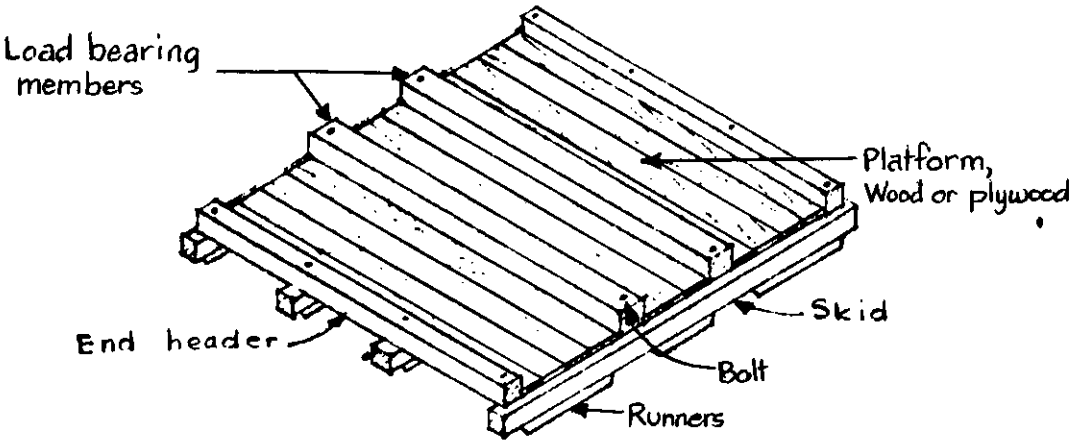




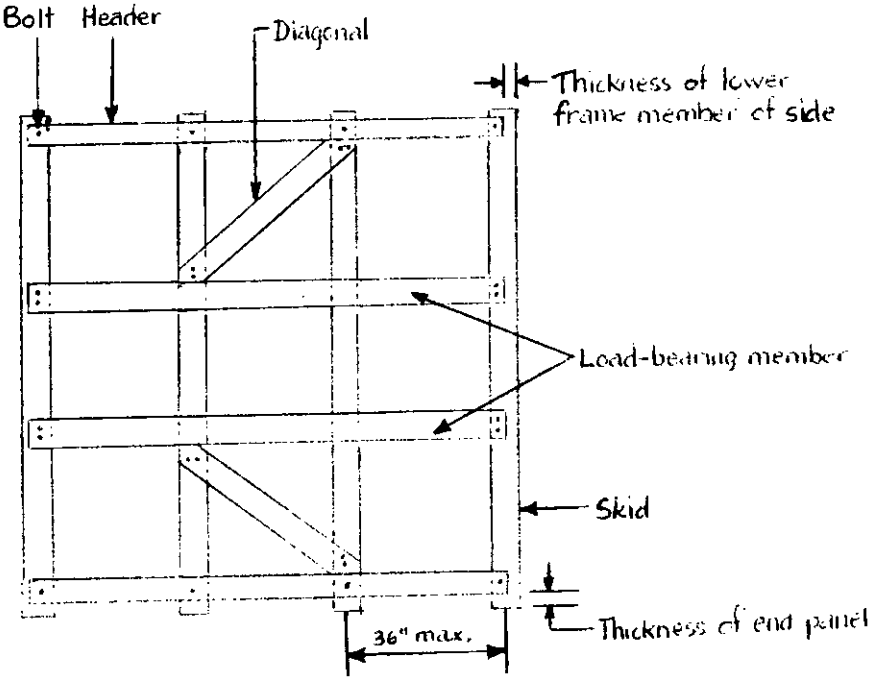
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Figure 10 - Open type crate covers for propellers 60 to 108 inches in diameter.

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COVERED TYPE SKID

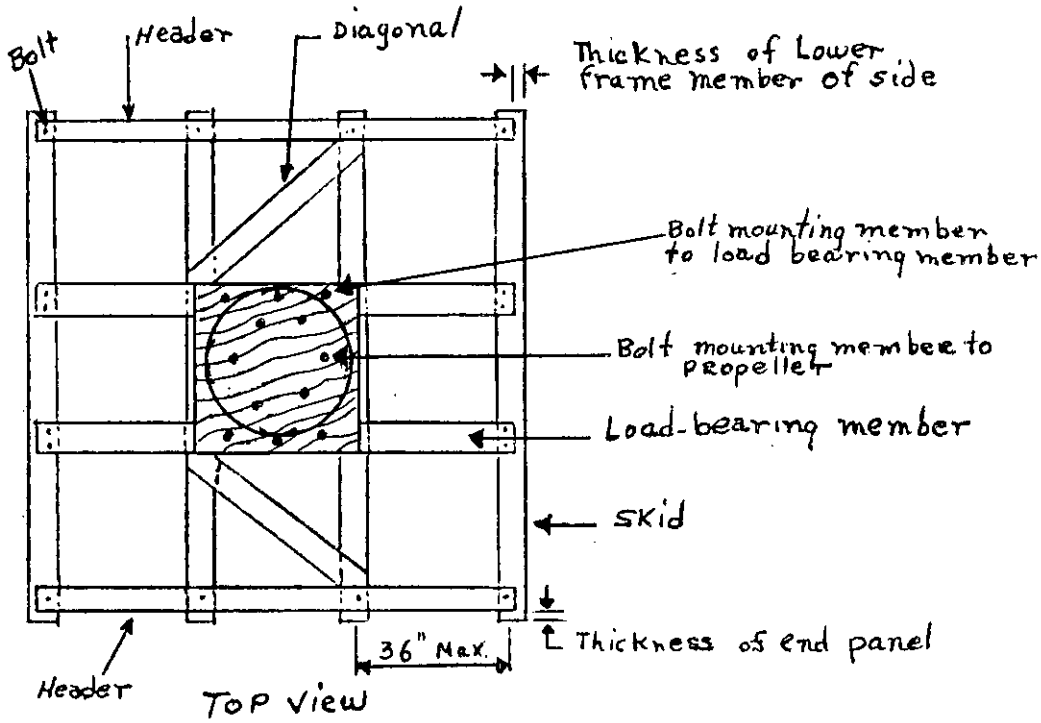


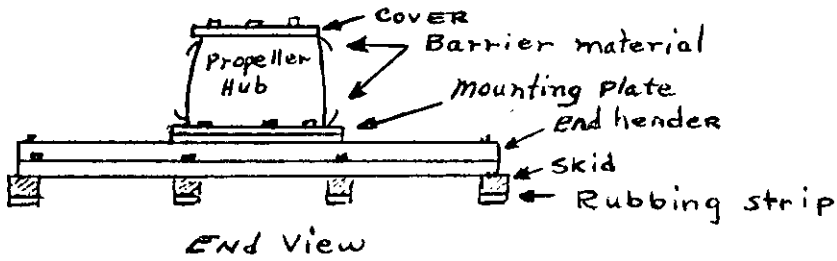
SH10549

OPEN TYPE SKID

Figure 11 - Skid type bases for propellers 60 to 108 inches in diameter.

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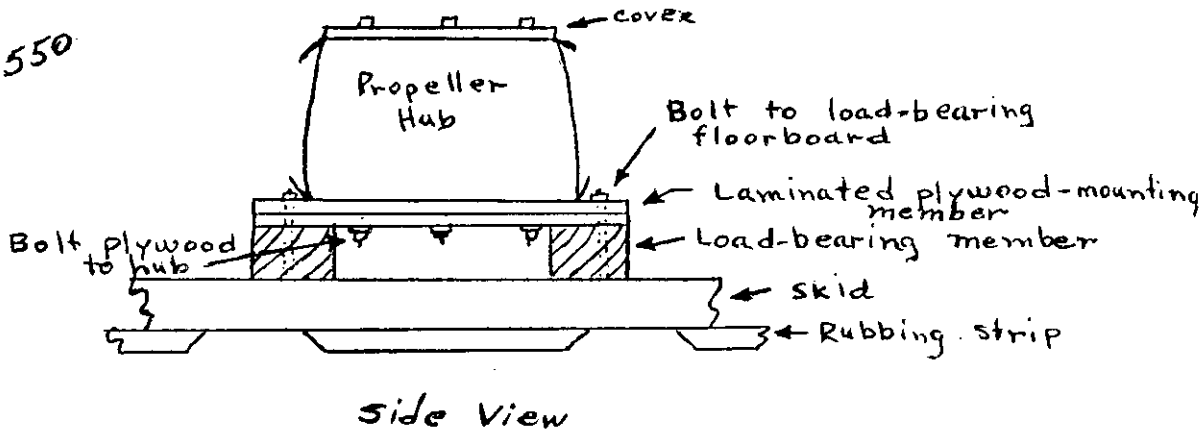


Figure 12 - Propeller anchoring.

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Figure 13 - Loading of propellers (more than 108 inches in diameter).

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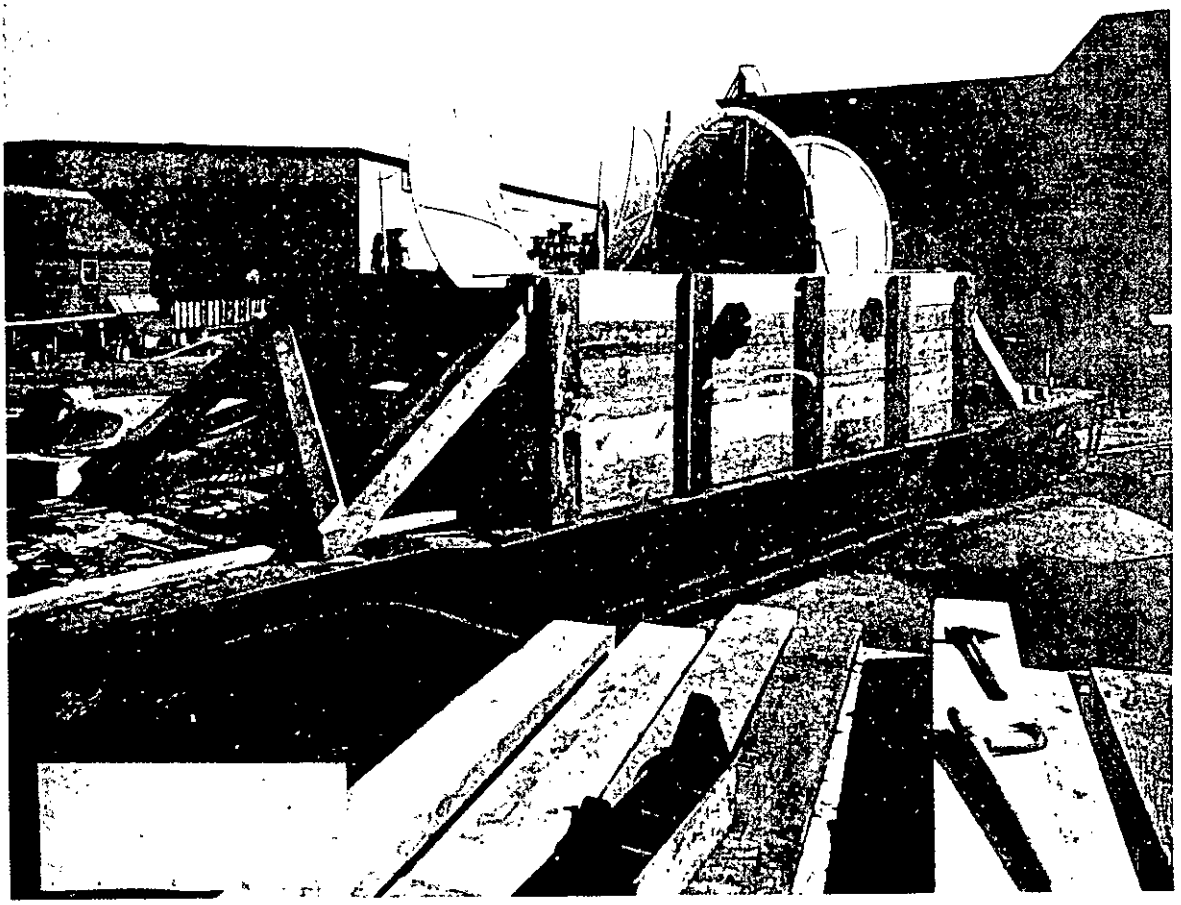
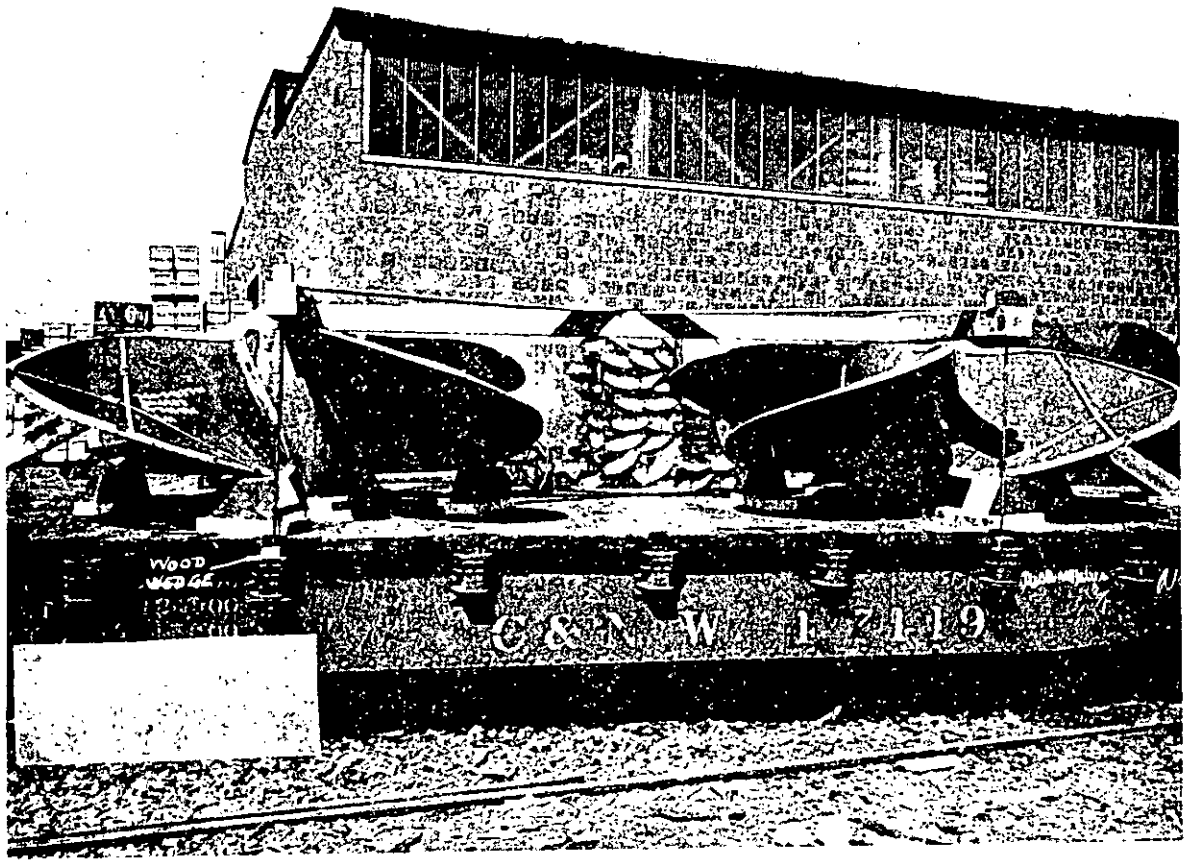


Figure 14 - Loading of propellers (more than 108 inches in diameter).

SH 8945

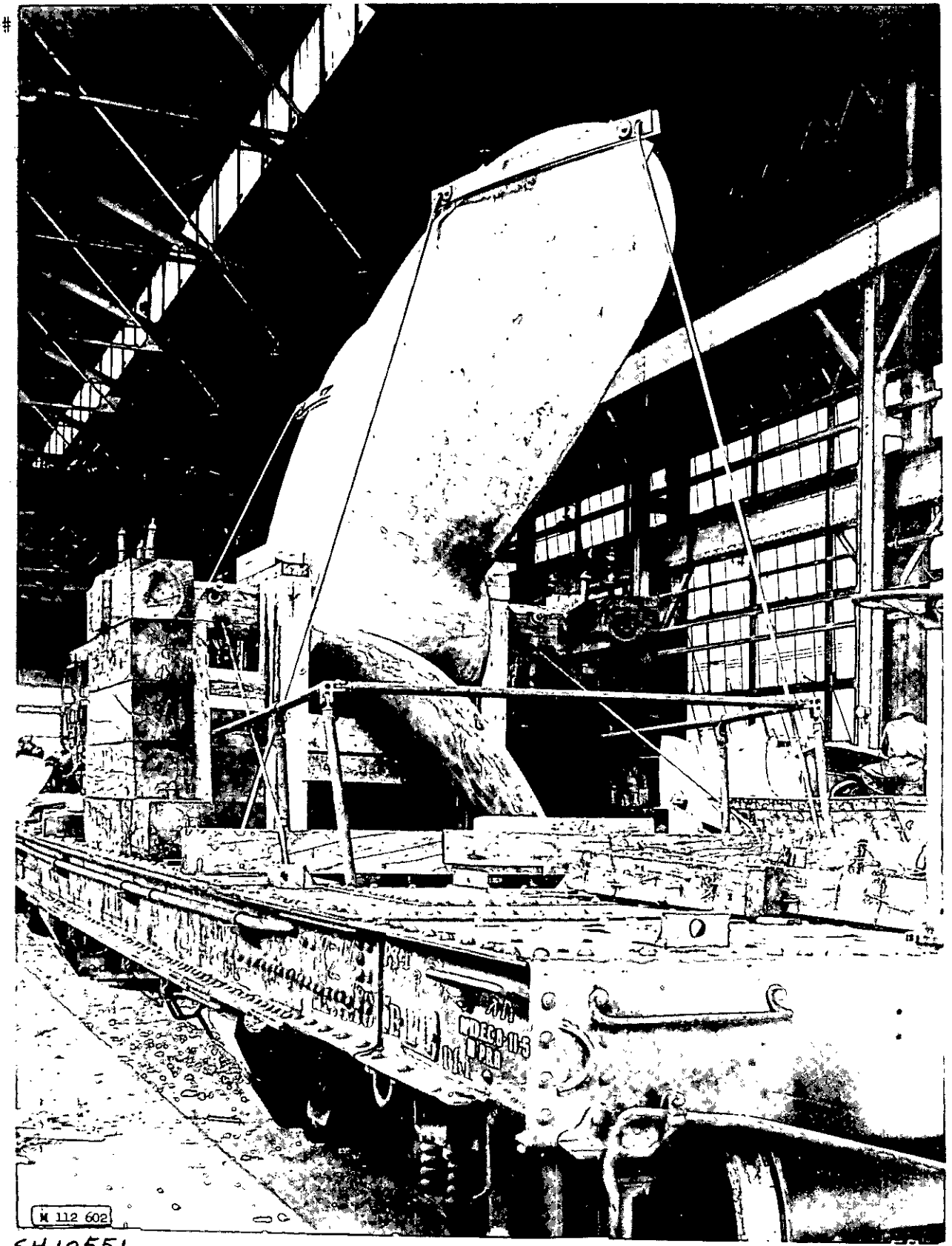
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Figure 15 - Vertical shipment of propellers (more than 108 inches in diameter).

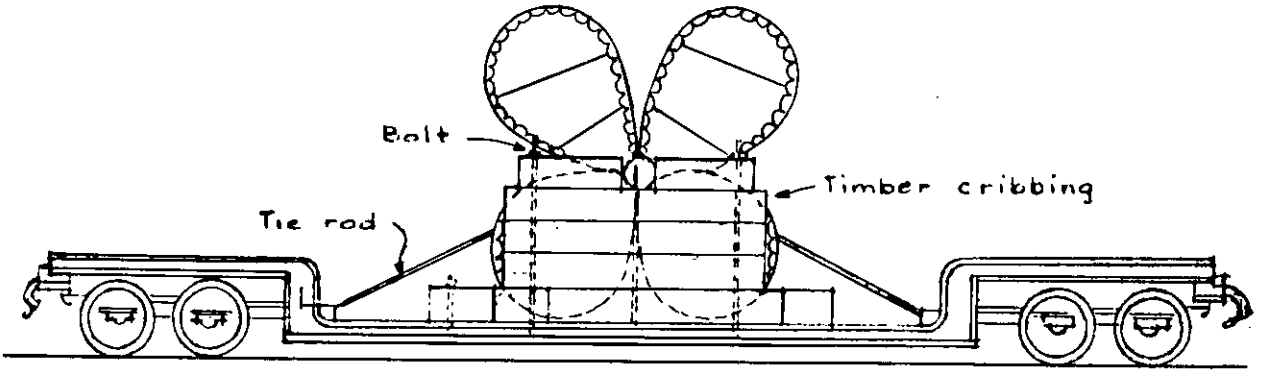
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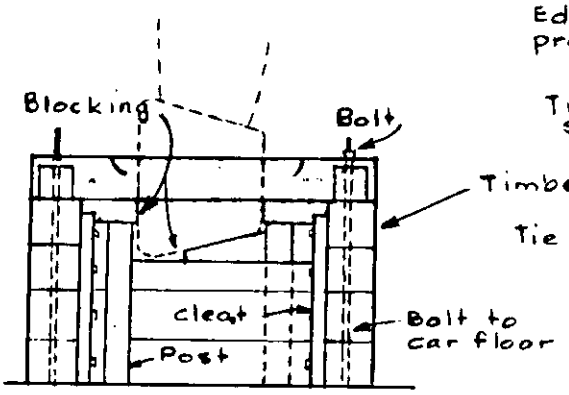
SH 10551

Figure 16 - Well car.

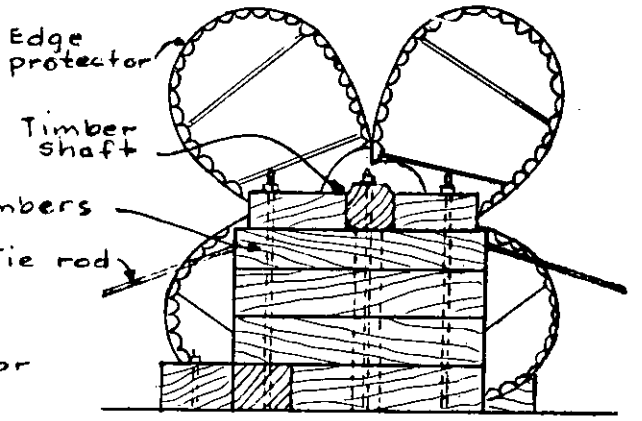
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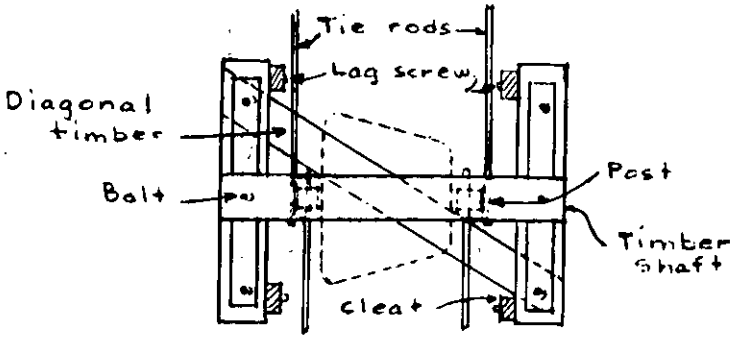
DROPPED-CENTER CAR



END VIEW



SIDE VIEW



SH10552

TOP VIEW

Figure 17 - Blocking of large propeller.

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Figure 18 - Propeller shipping frame.

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